

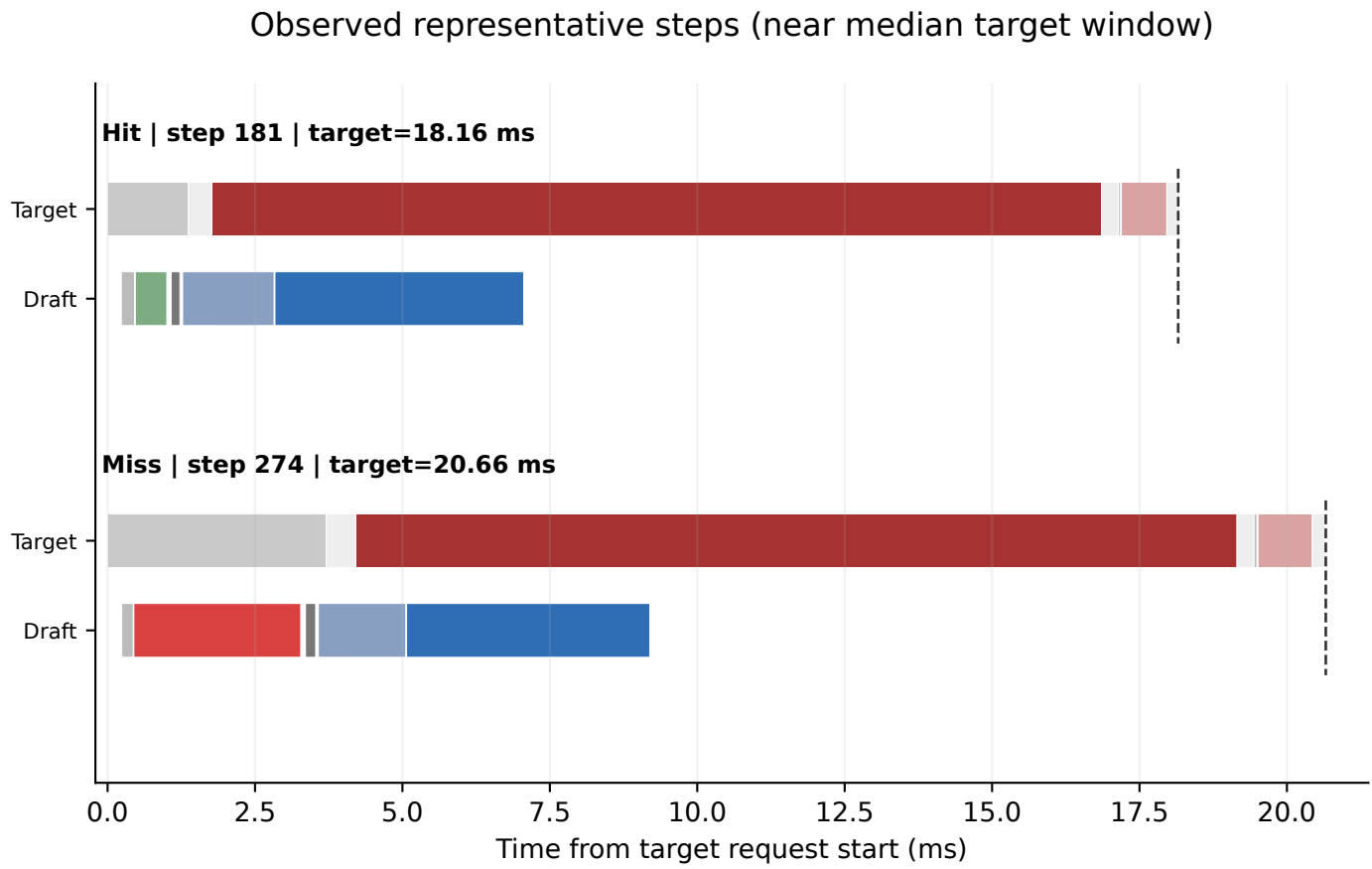
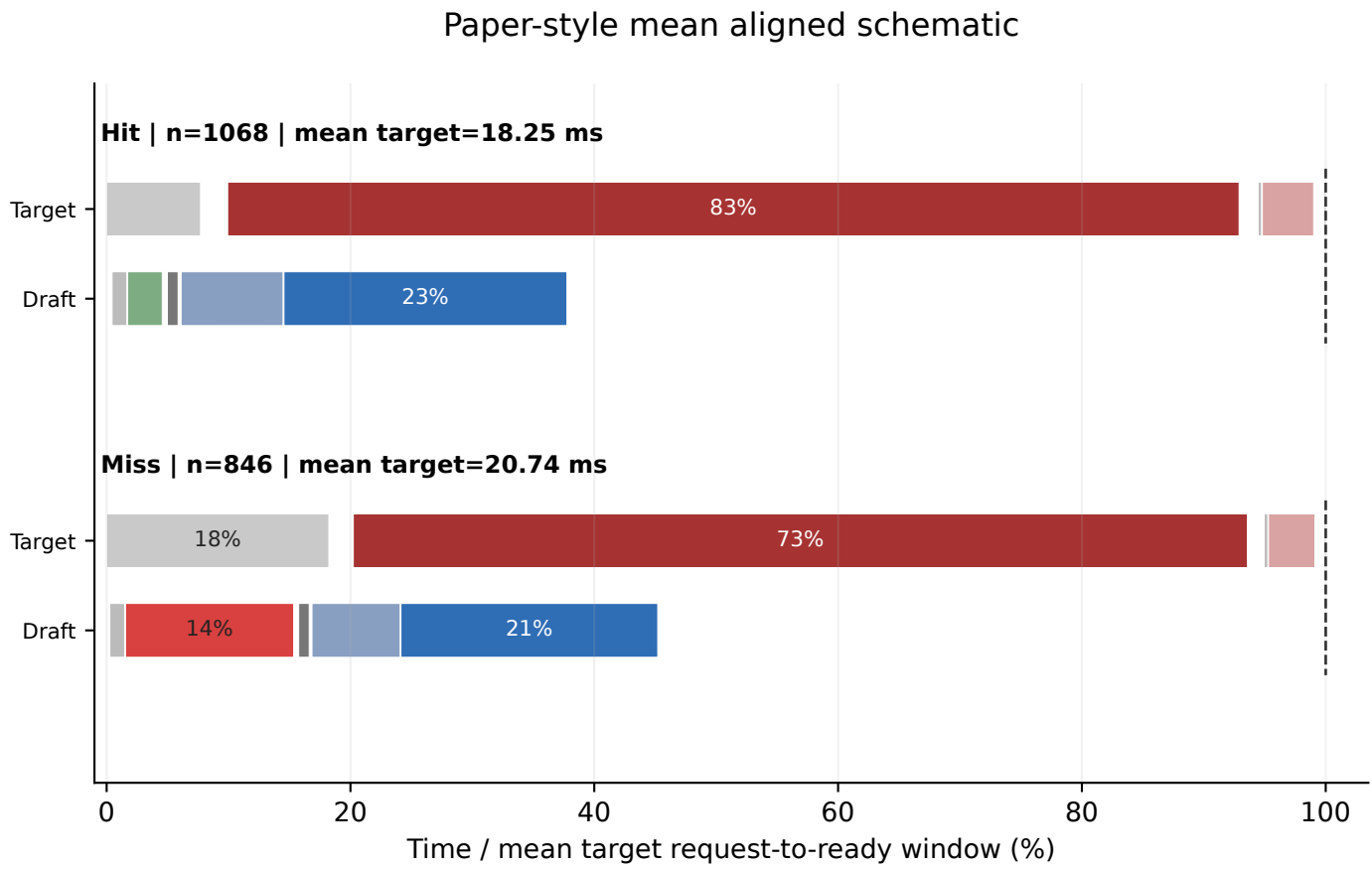
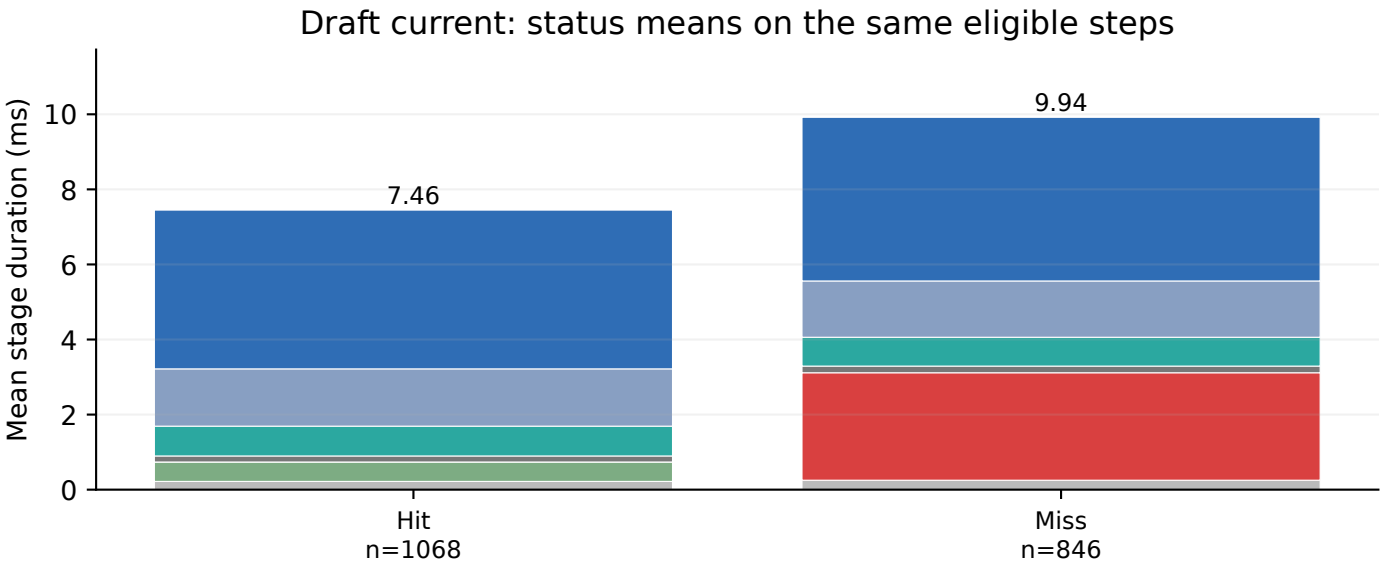
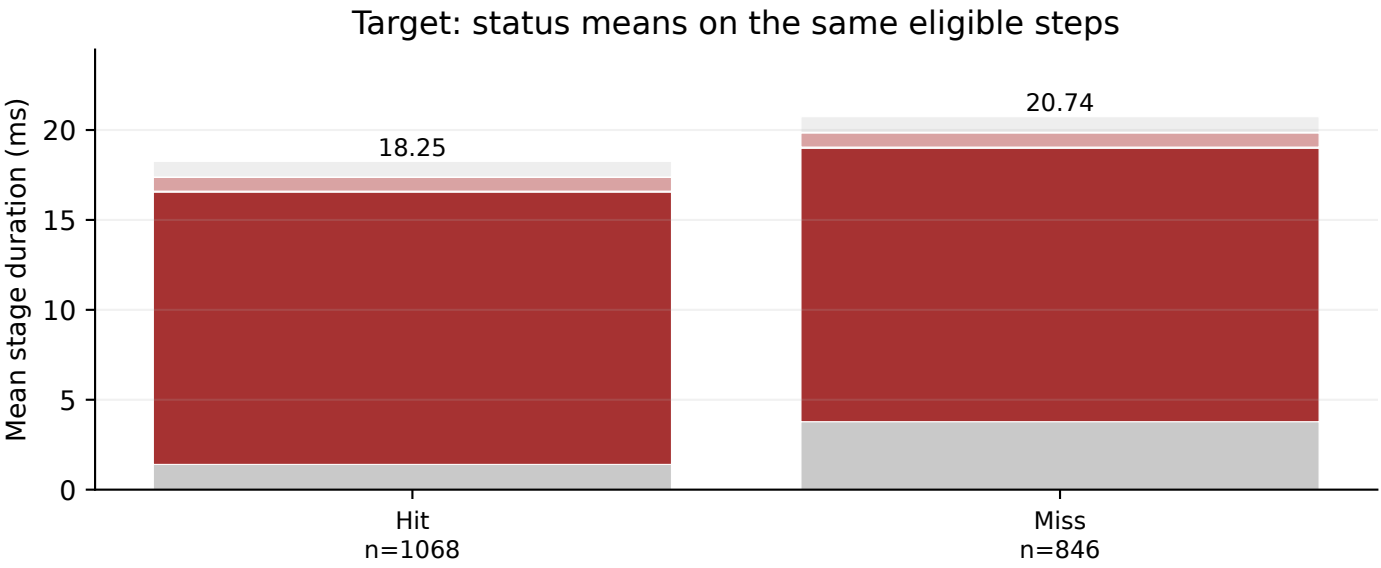
Final selected configurations: DUET and independently tuned SSD

Model	B	Method	AL*	Cache hit	TPS* r0 / r1
LLAMA2	1	SSD	2.143	54.9%	116.73 / 111.91
LLAMA2	1	DUET	2.346	84.8%	127.03 / 126.84
LLAMA2	8	SSD	2.151	54.4%	683.09 / 681.79
LLAMA2	8	DUET	2.107	77.0%	604.86 / 597.28
LLAMA3	1	SSD	2.661	65.5%	123.40 / 121.30
LLAMA3	1	DUET	2.690	67.9%	115.88 / 117.21
LLAMA3	8	SSD	2.687	35.5%	684.36 / 694.42
LLAMA3	8	DUET	2.284	82.7%	537.68 / 542.47

Full 480 prompts | T=0.7 | RTX 4090 | input/output cap 512/128

AL and hit pool event counts across two repeats; TPS is shown separately for each paired repeat.

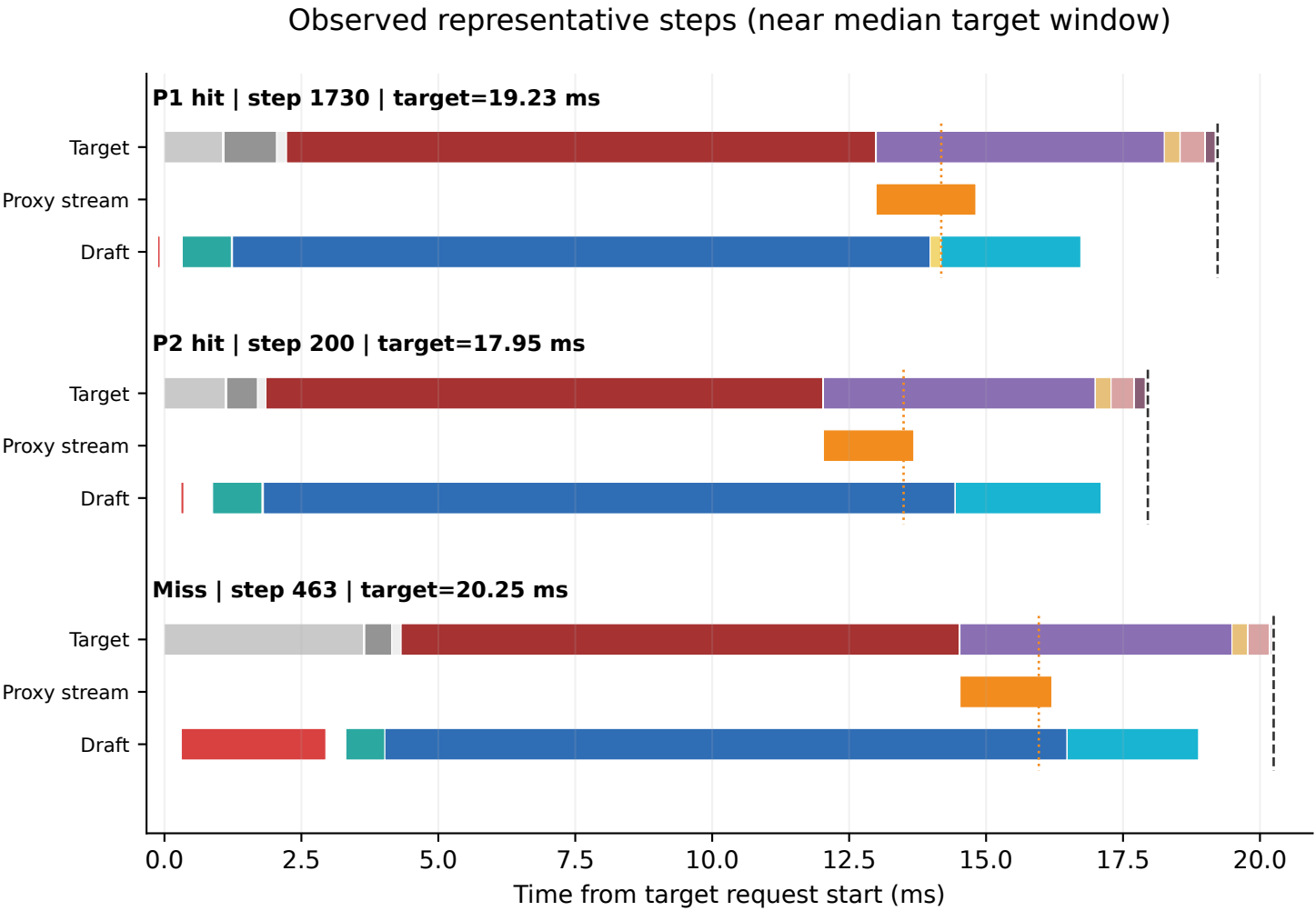
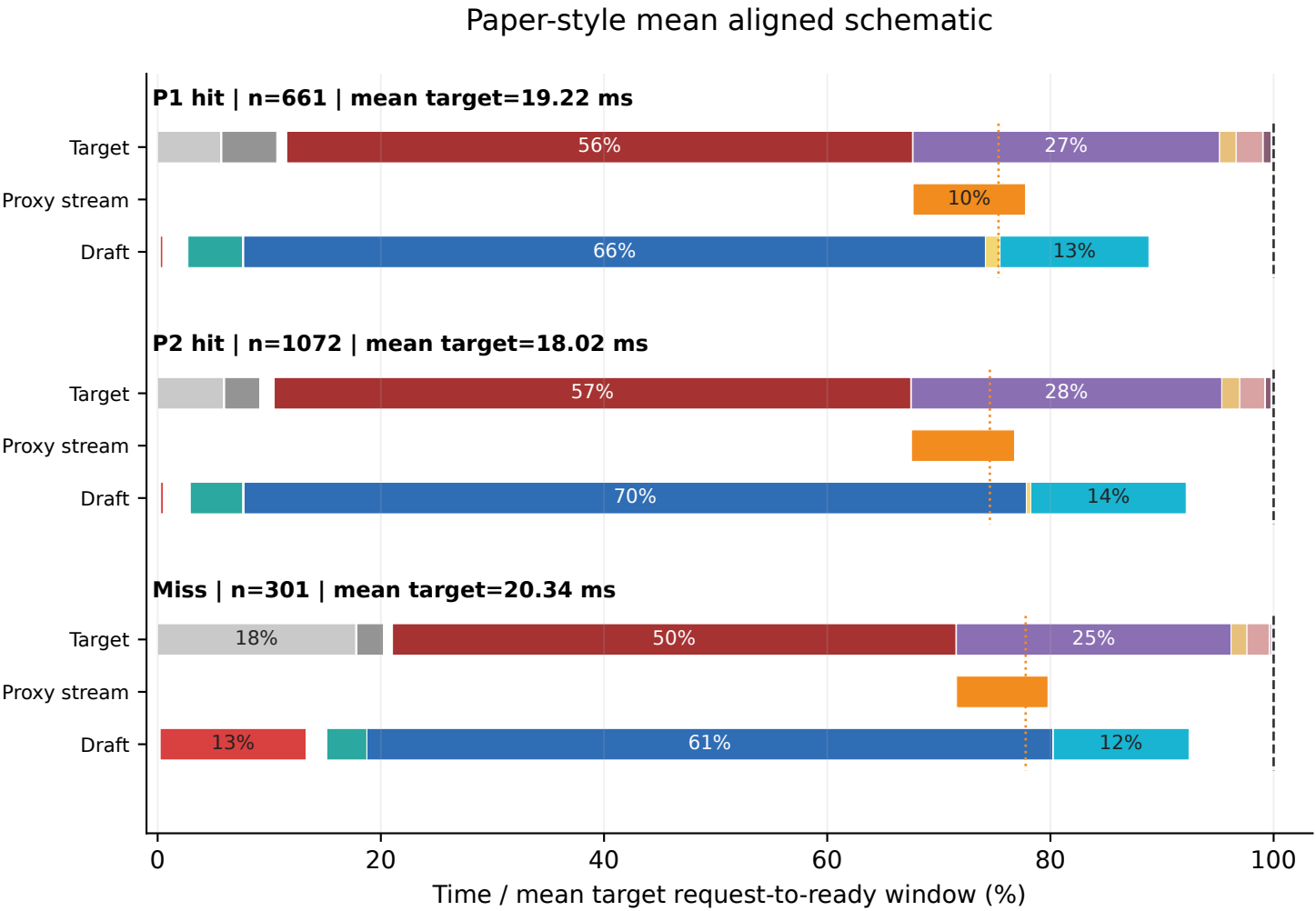
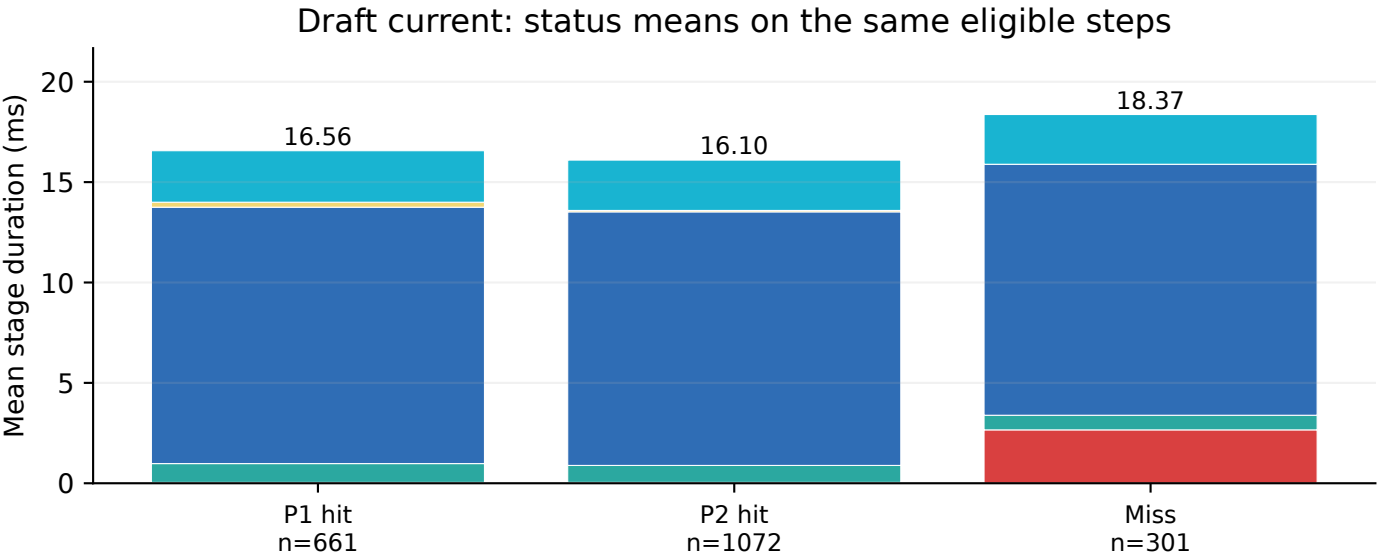
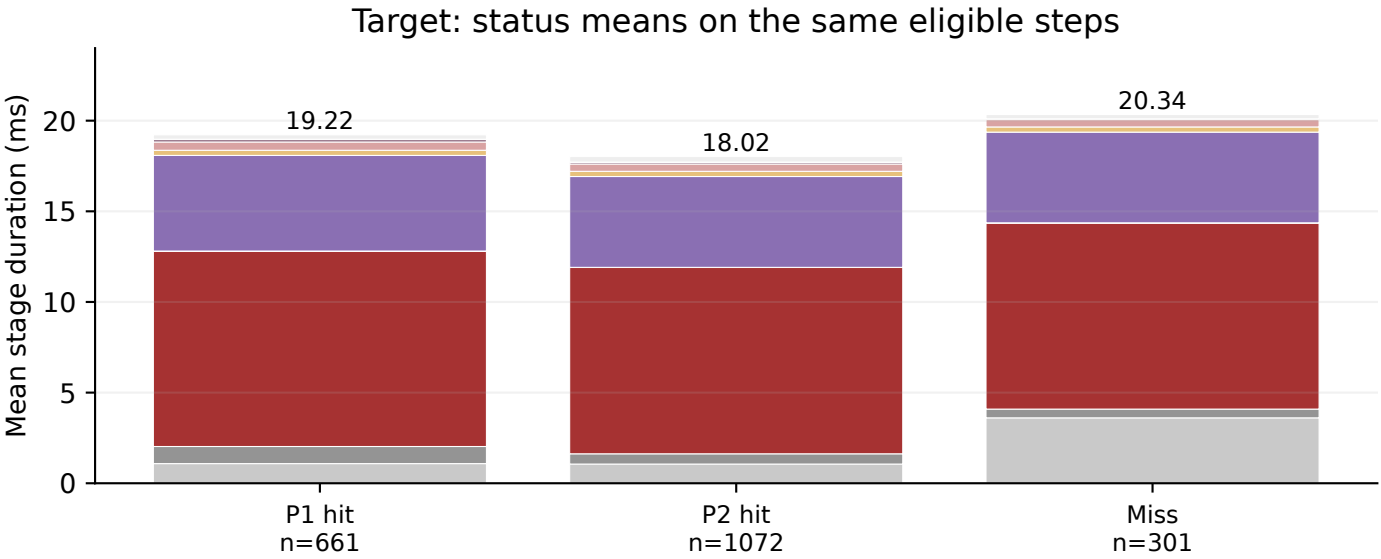
* Output-cap/clipping boundary exclusions. Raw returned TPS is preserved in the accompanying tables.



Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

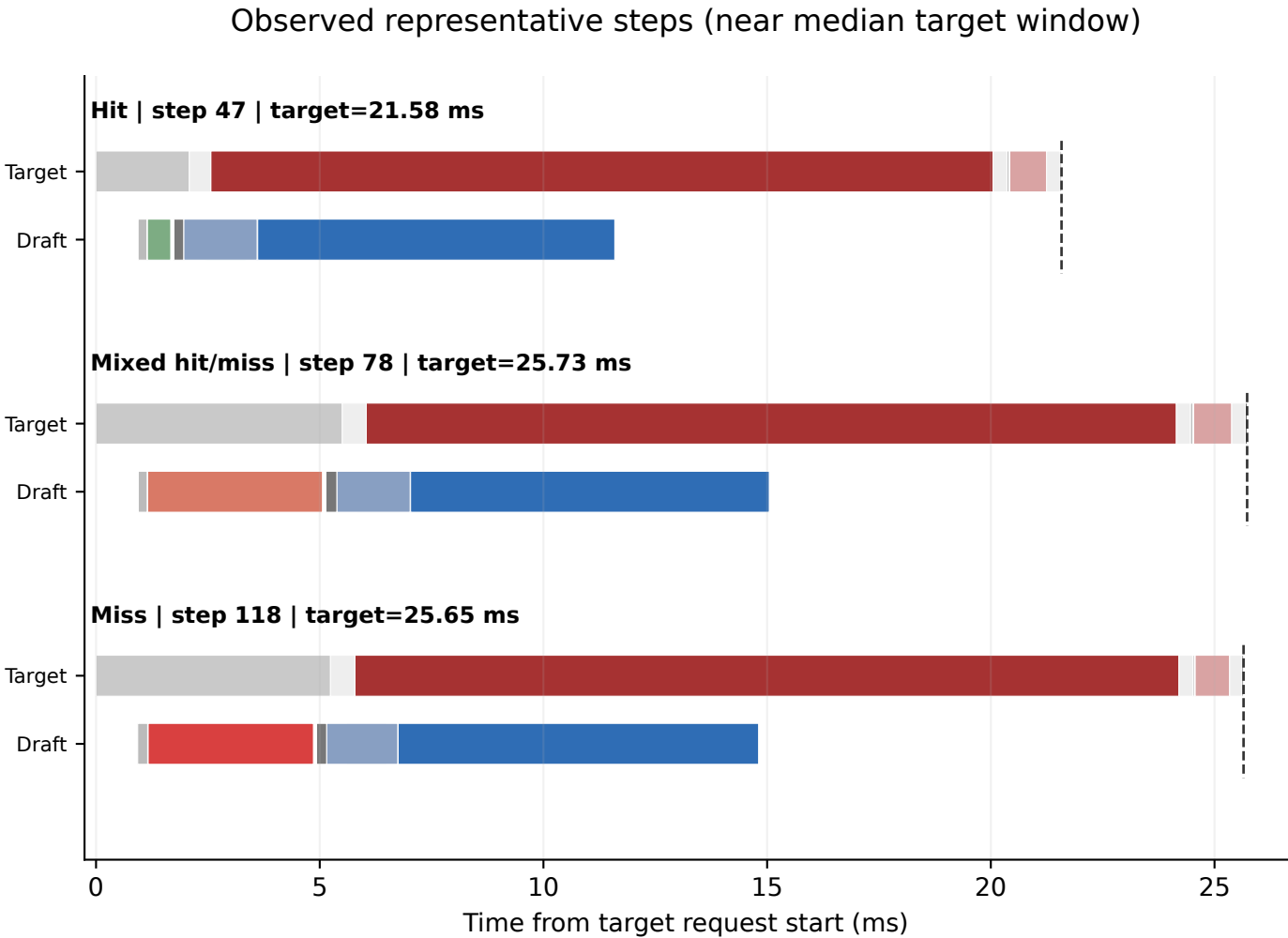
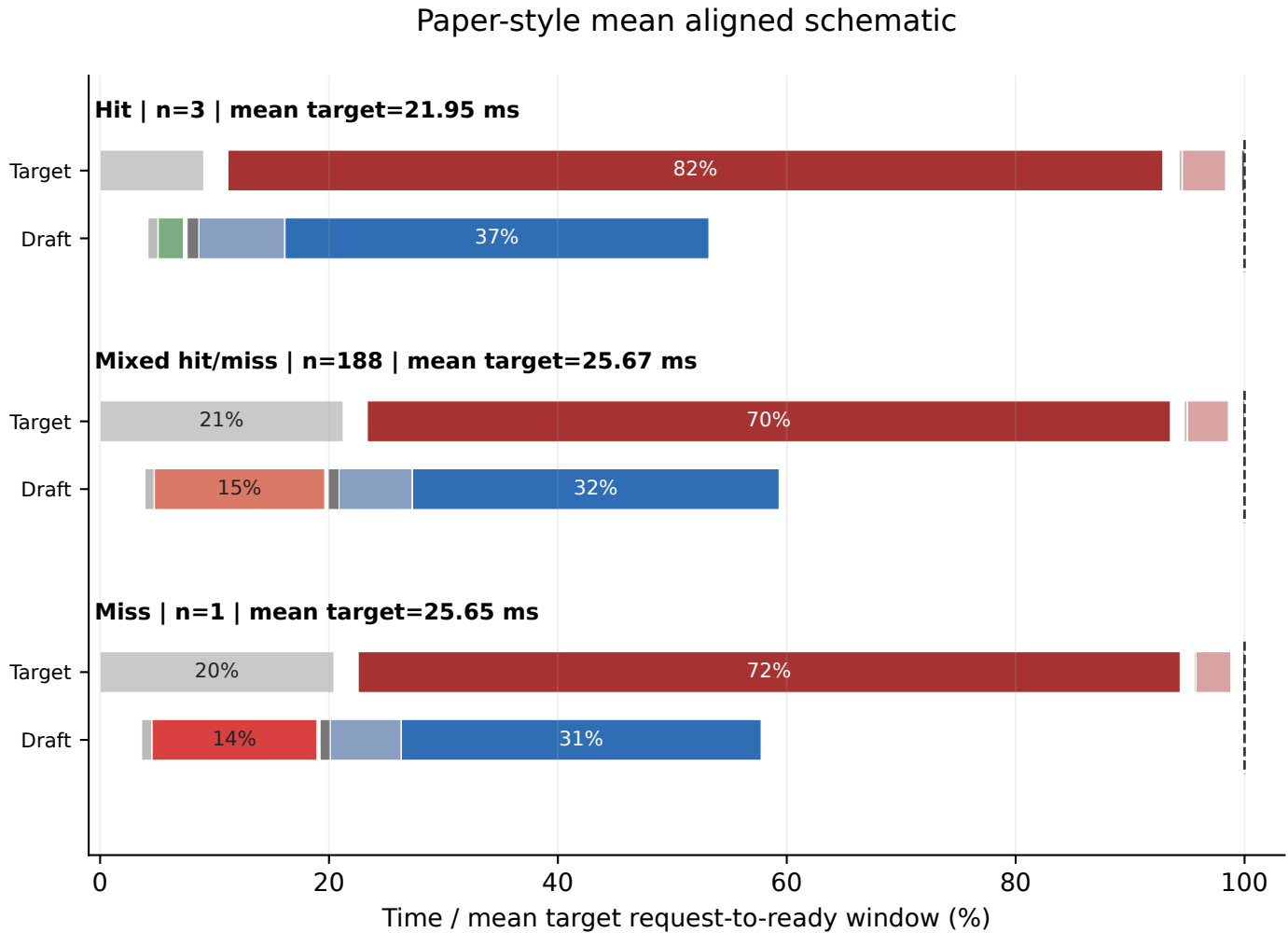
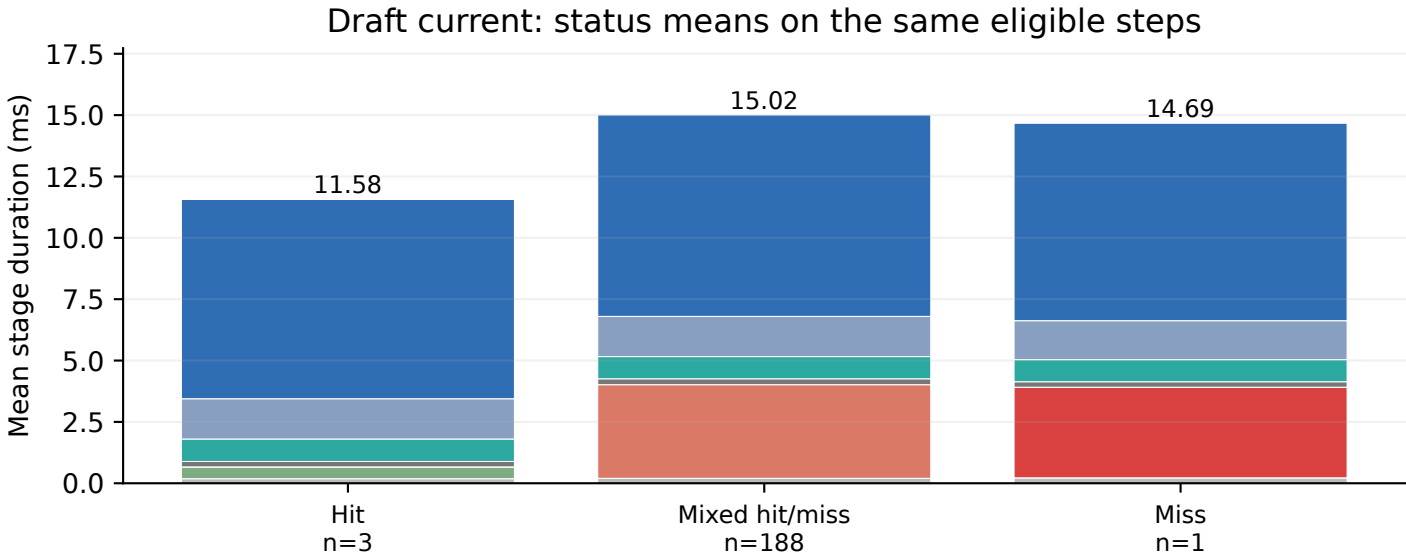
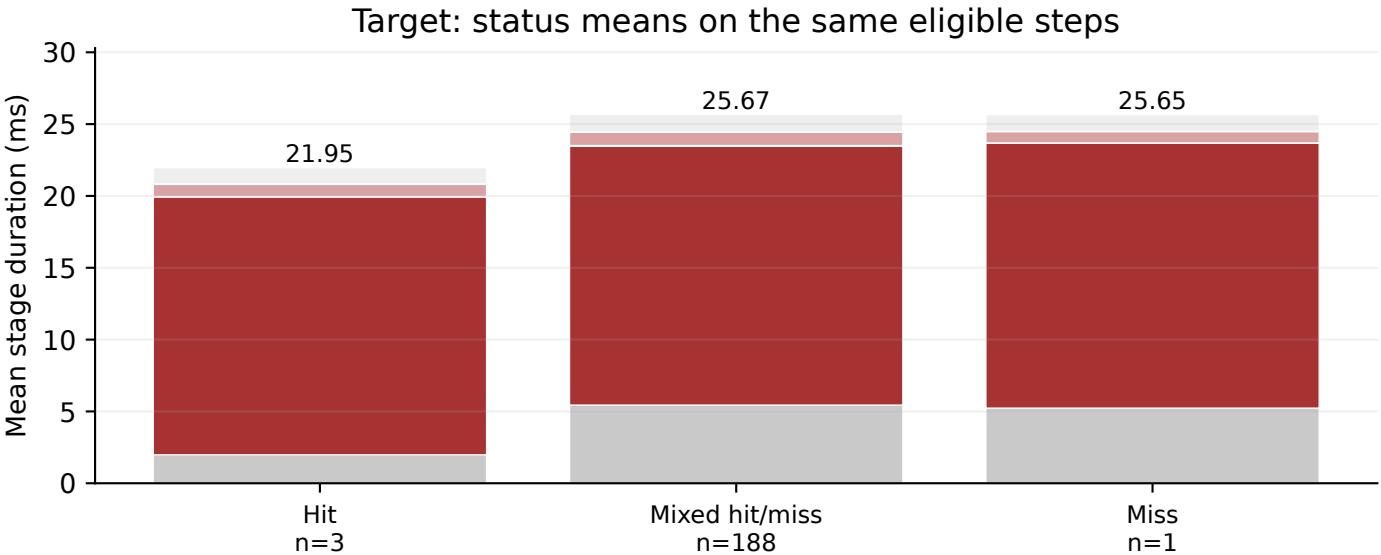
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_postopt_profile_k1_down
DUET: K1/K2=11/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=12, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 2034 eligible steps



Wait / sync Target post Final logits Postprocess Context P2 Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Proxy wait Proxy received
Target pre Proxy compute Commit Miss draft

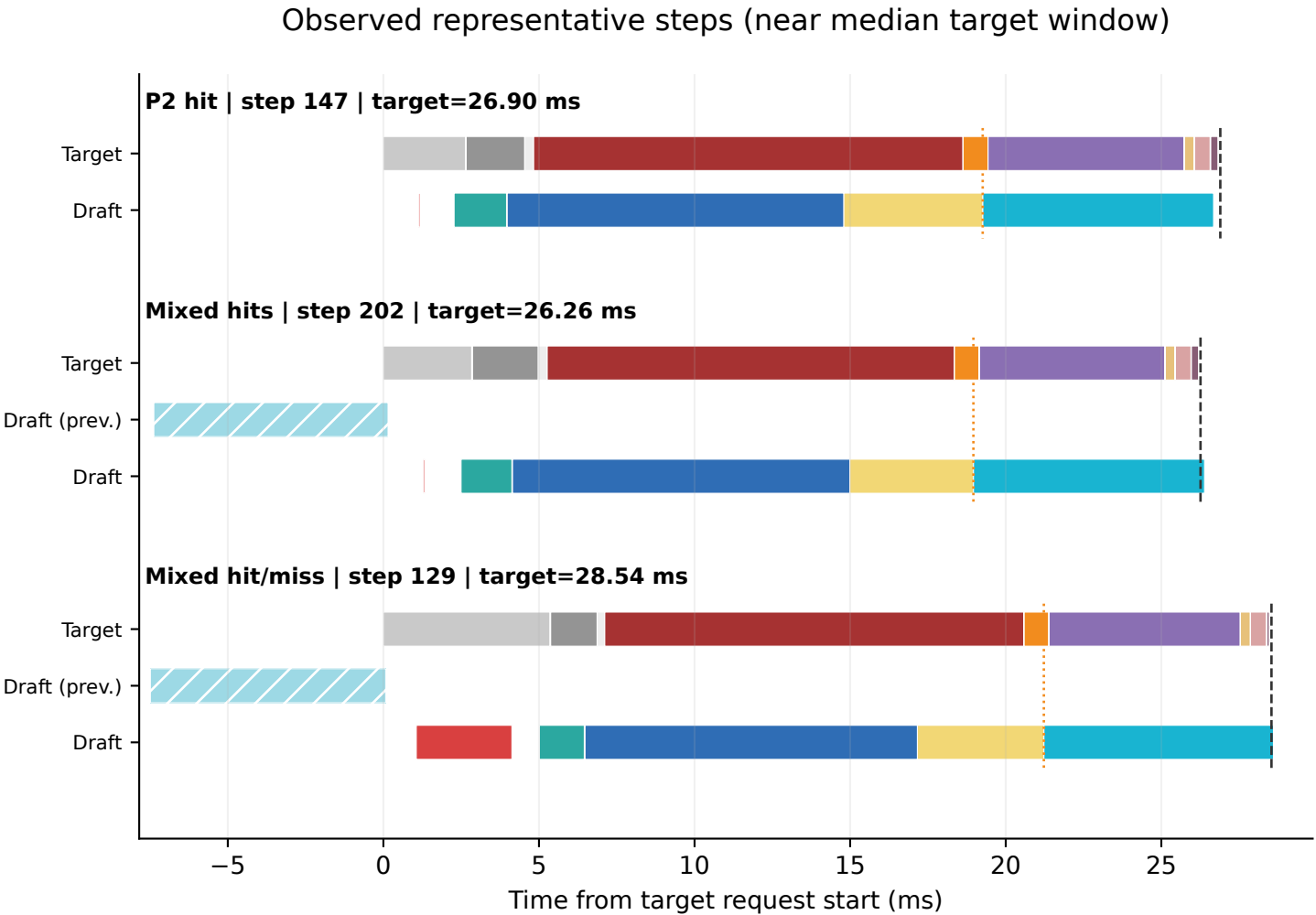
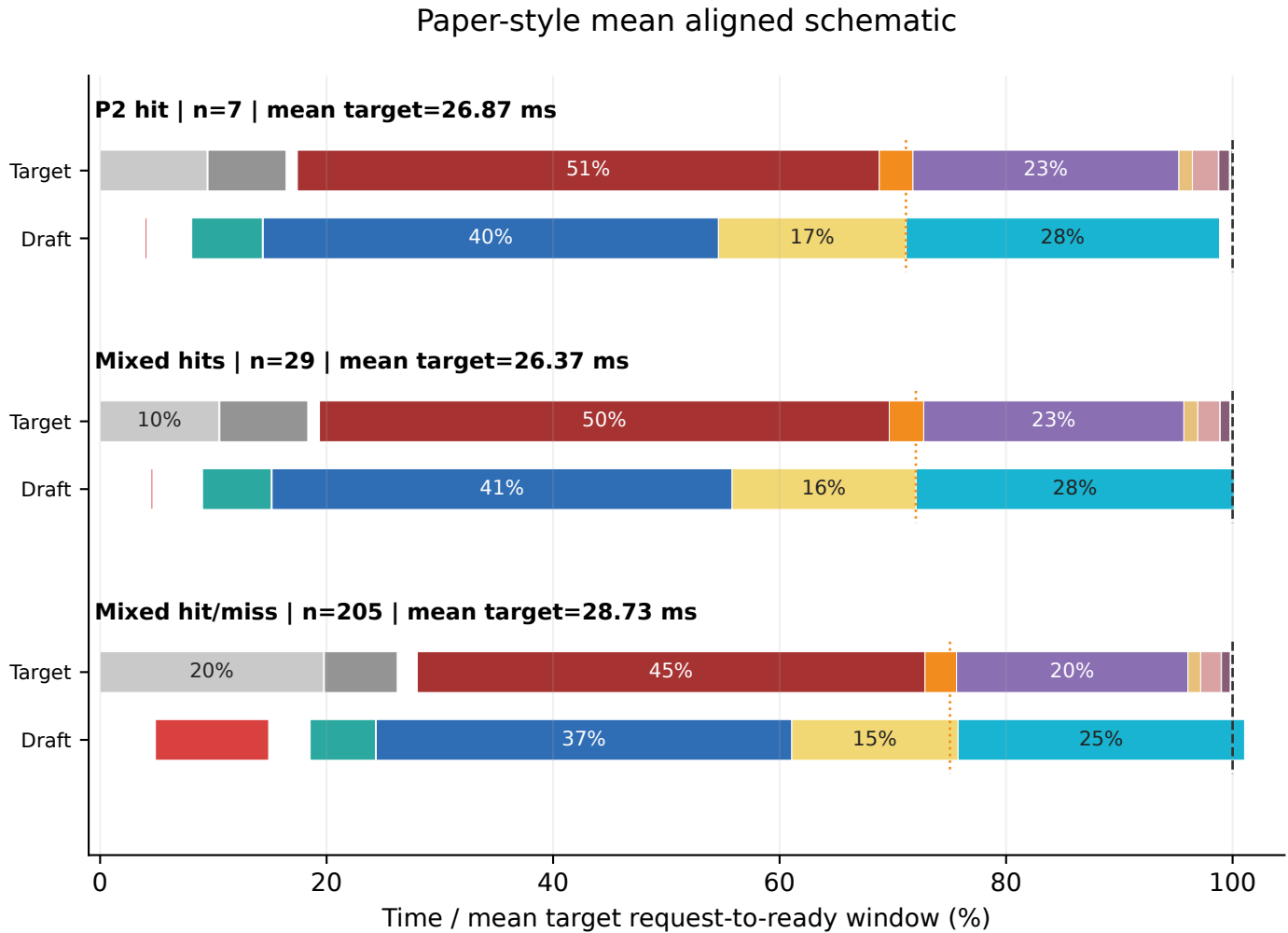
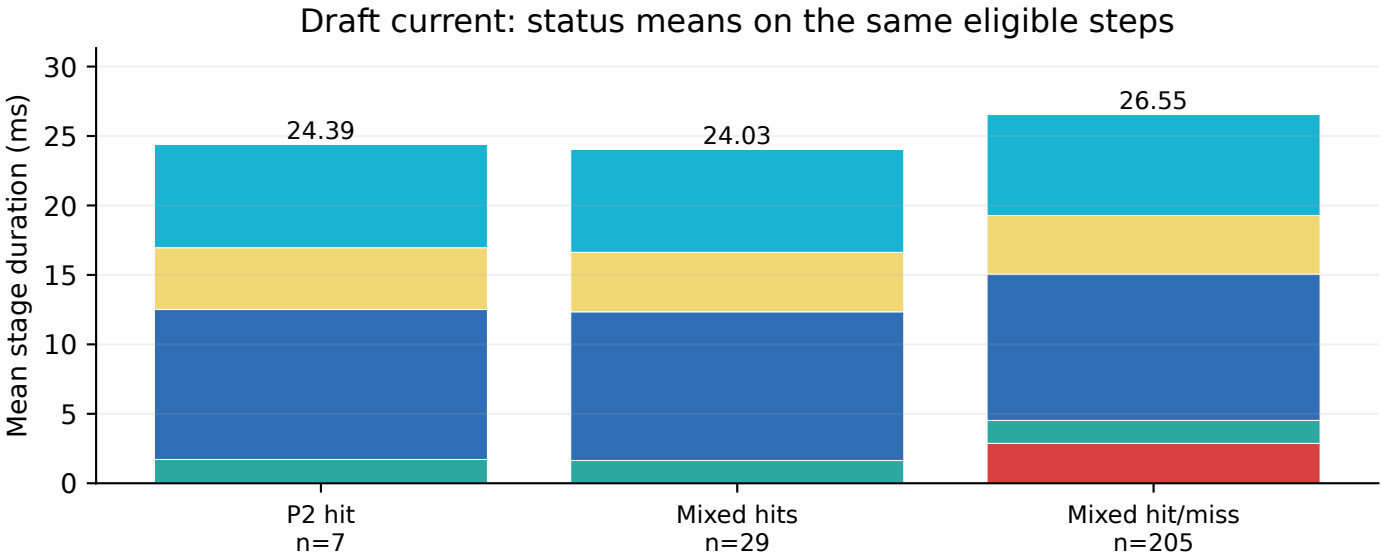
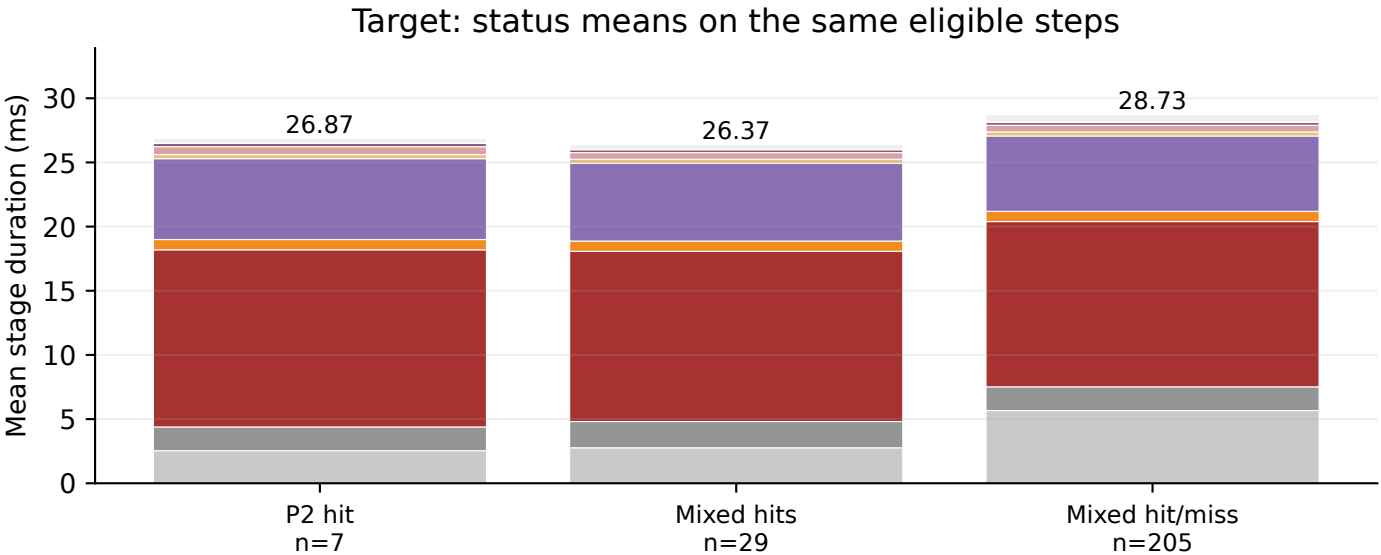
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Mixed response Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap Context

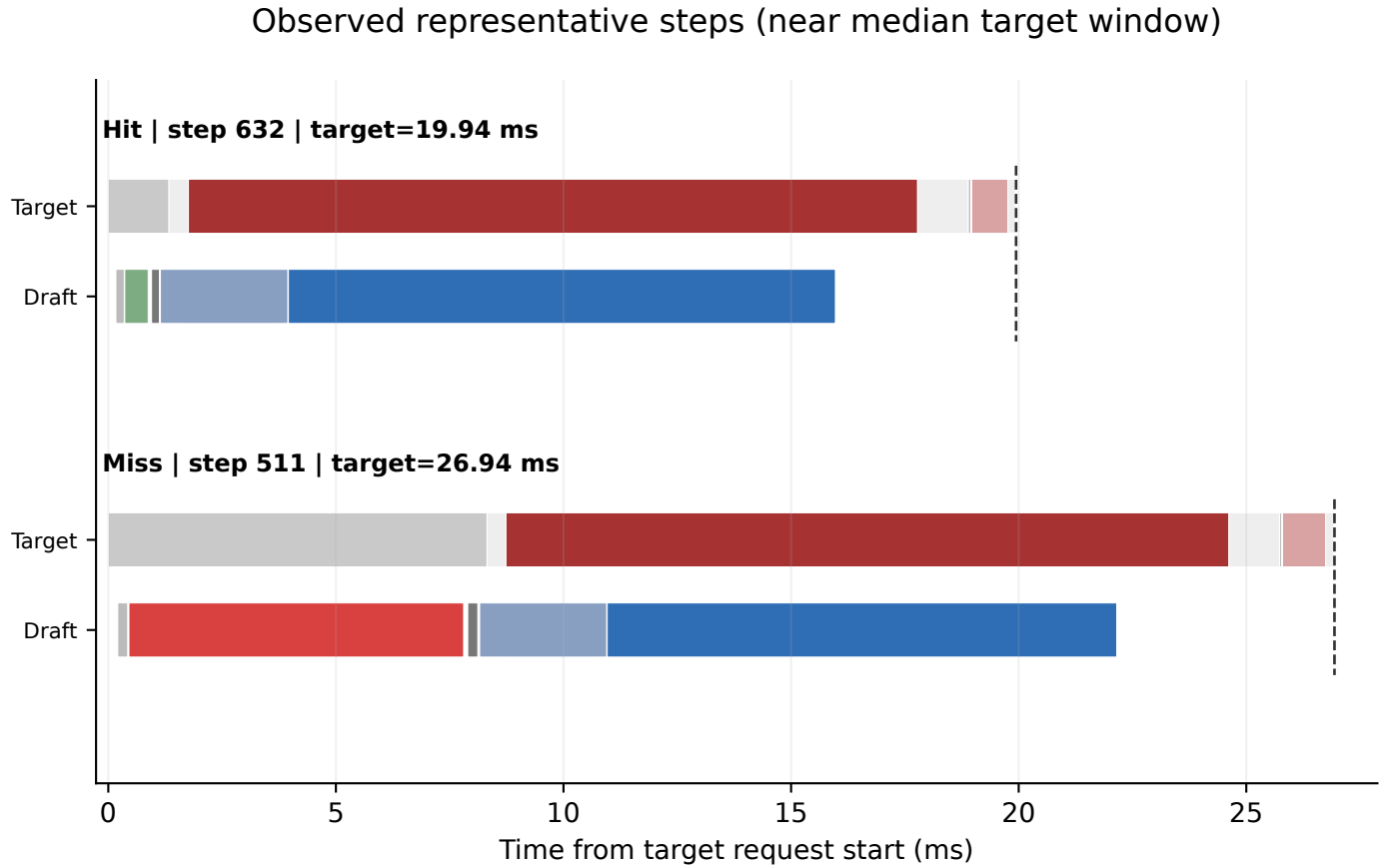
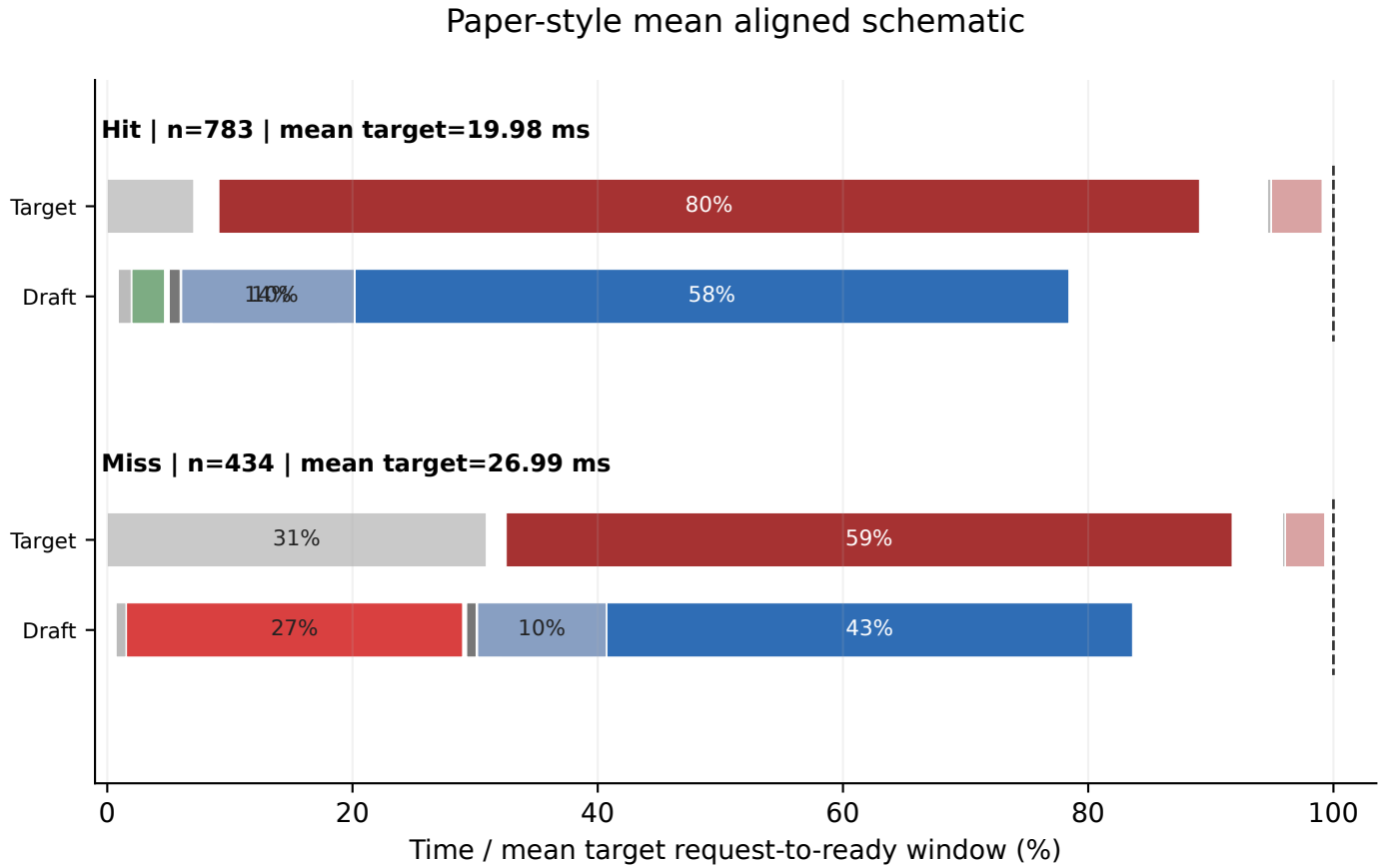
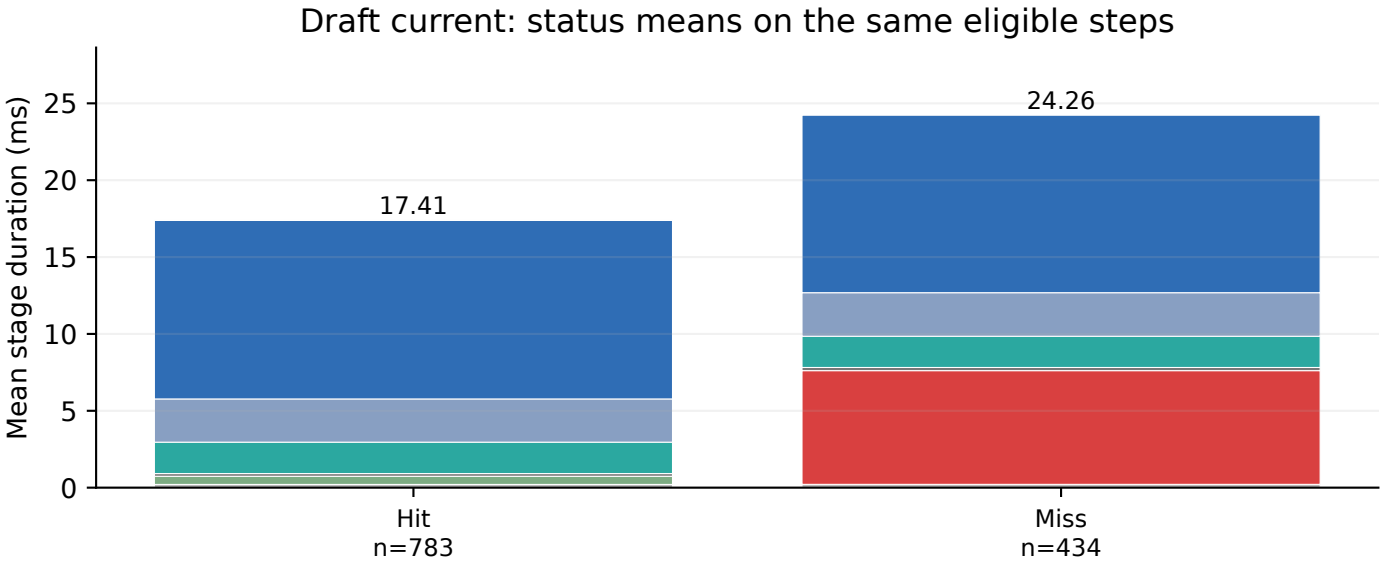
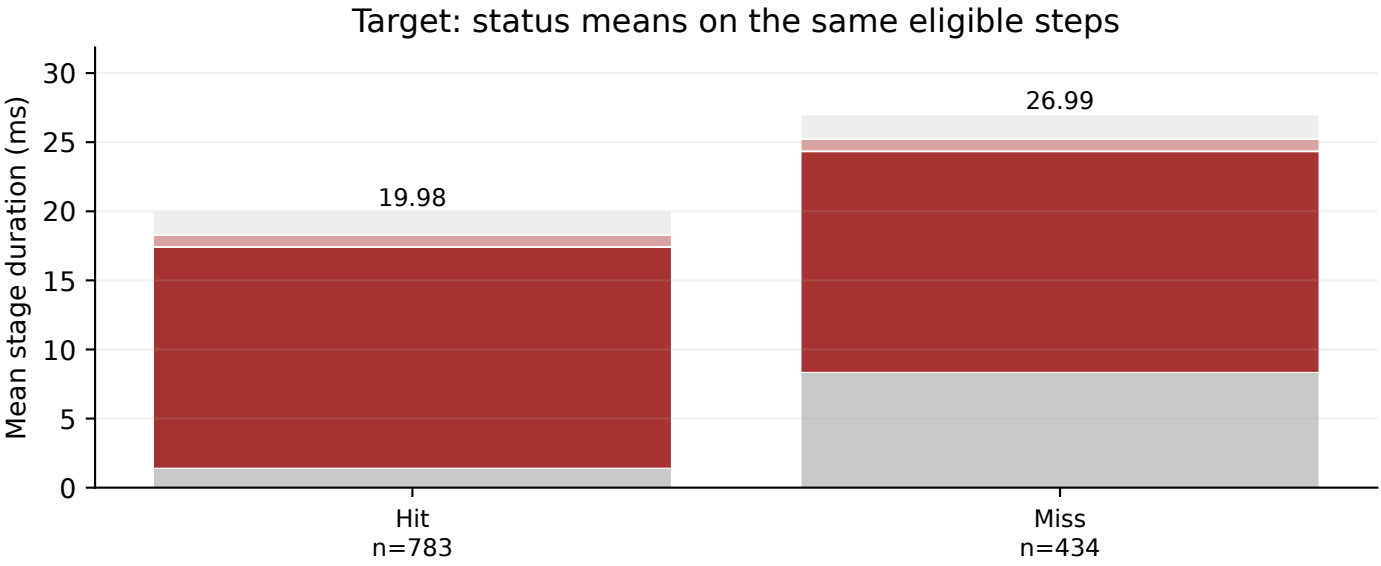
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_opt_l0_profile_trim1
DUET: K1/K2=3/2, exit=21 (0-based), C=3, N1/N2=6/4, U=1, W=4, trim/stream=1/0
B=8, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 241 eligible steps



Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received
Target pre Final logits Postprocess Context

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

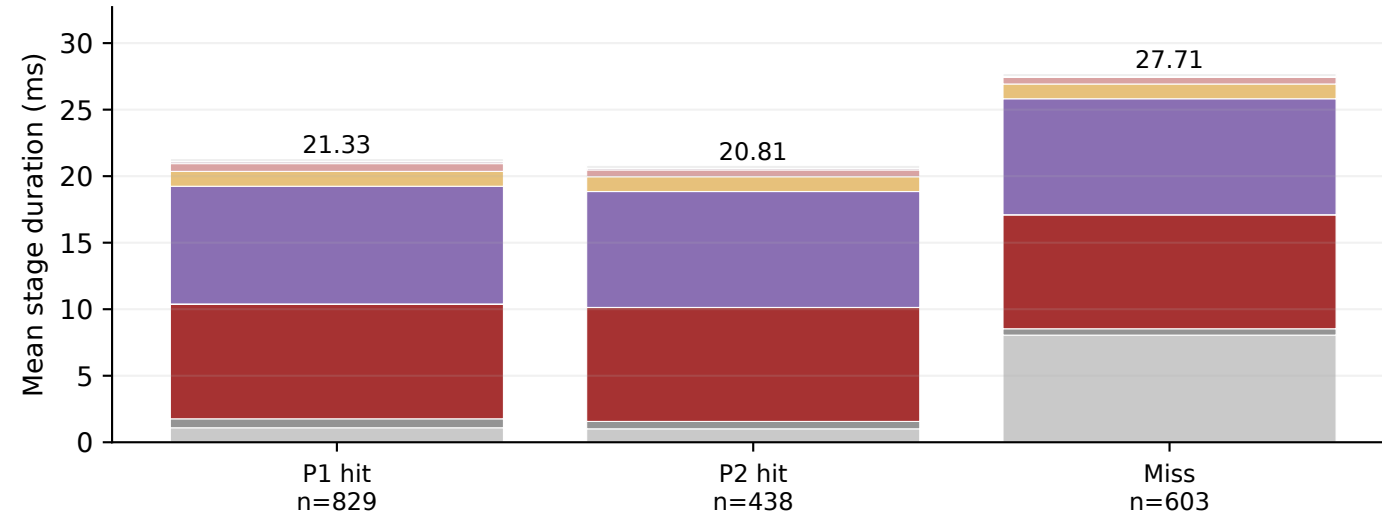


Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

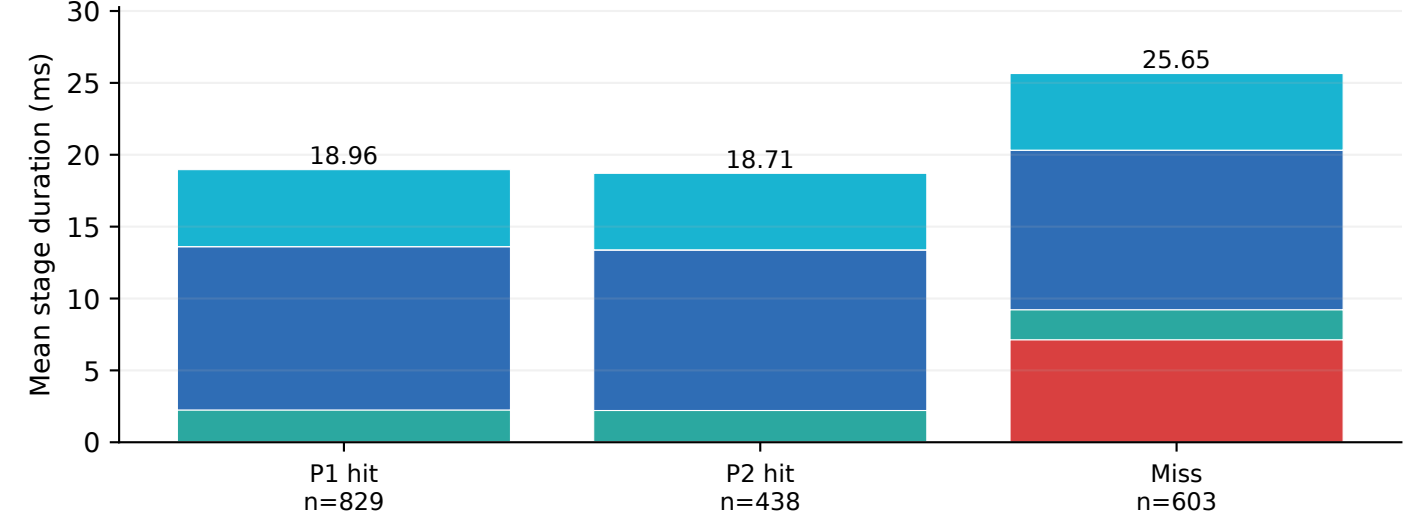
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_stream_l0_profile_s1
DUET: K1/K2=4/2, exit=16 (0-based), C=3, N1/N2=8/4, U=2, W=5, trim/stream=1/1
B=1, T=0.7, 48 prompts, cap=512/128 | instrumented diagnostic; 1870 eligible steps

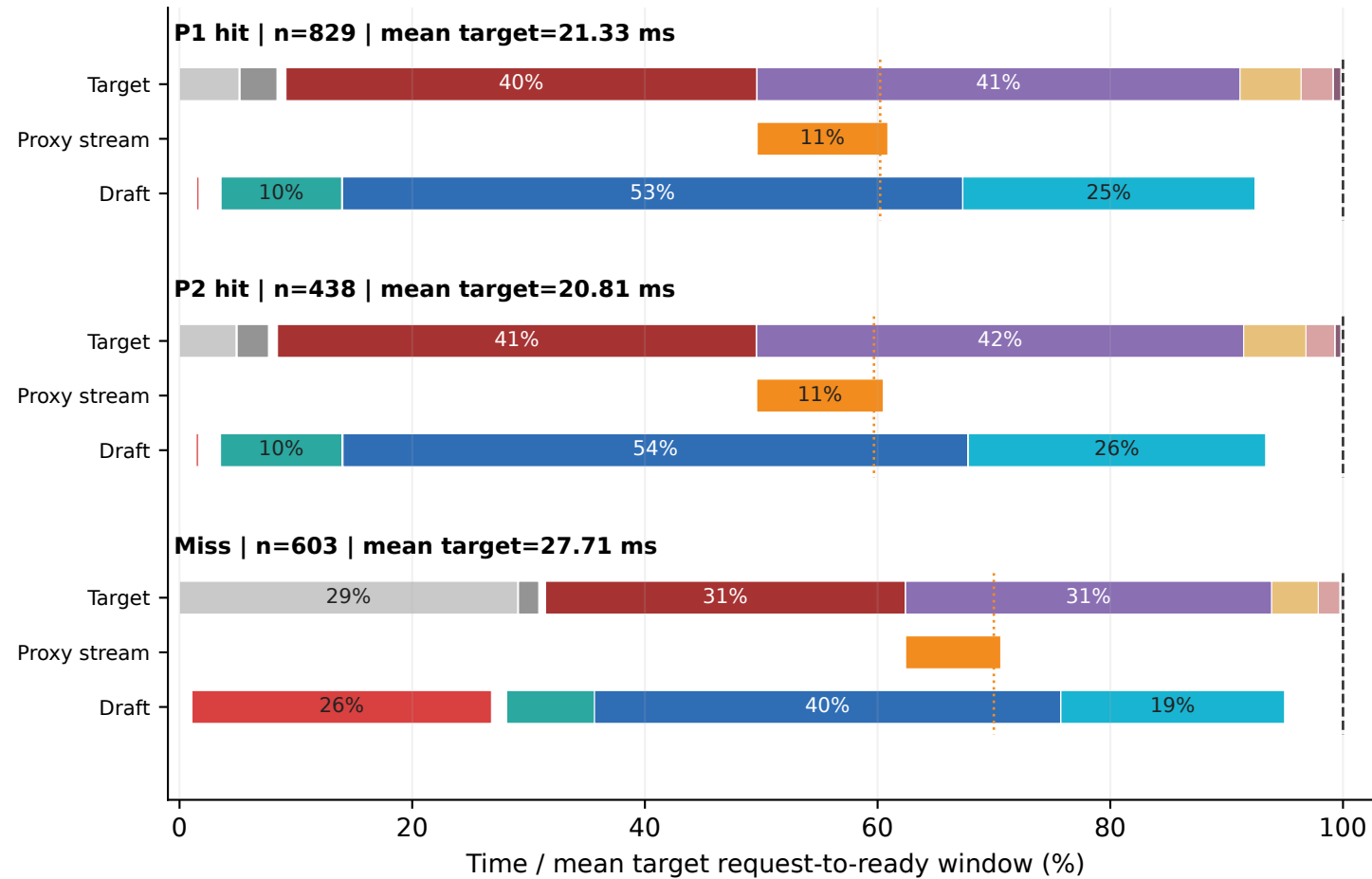
Target: status means on the same eligible steps



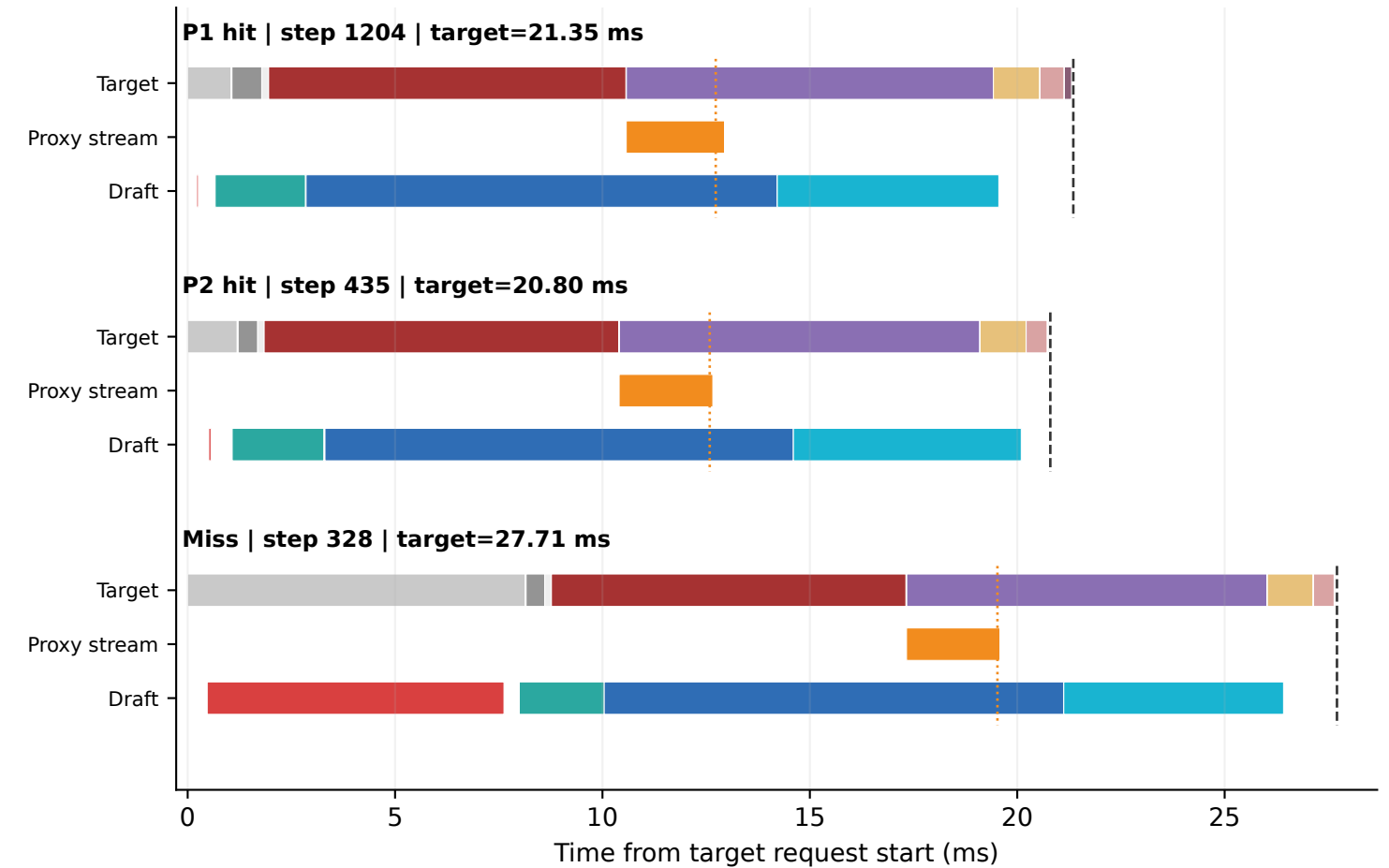
Draft current: status means on the same eligible steps



Paper-style mean aligned schematic

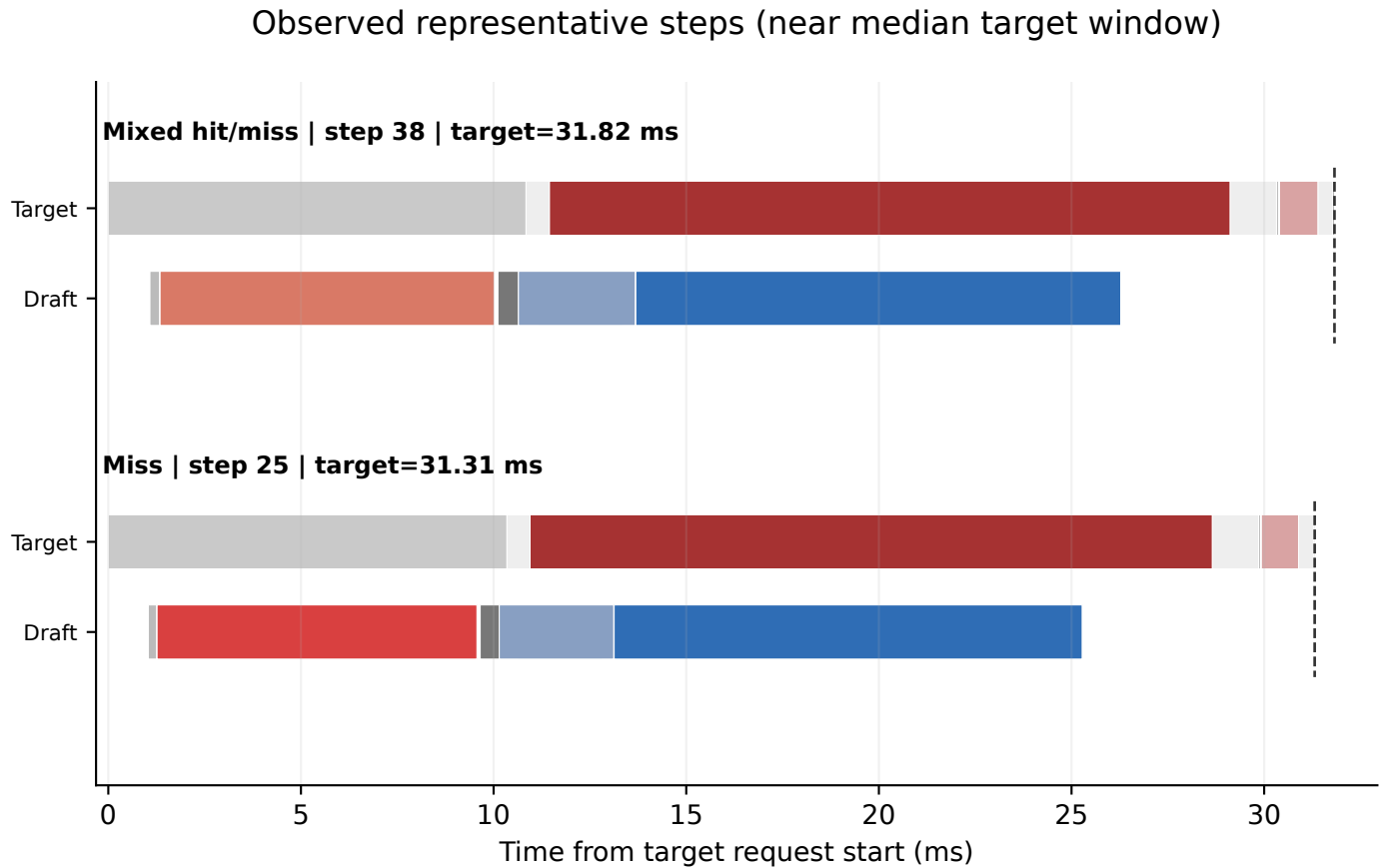
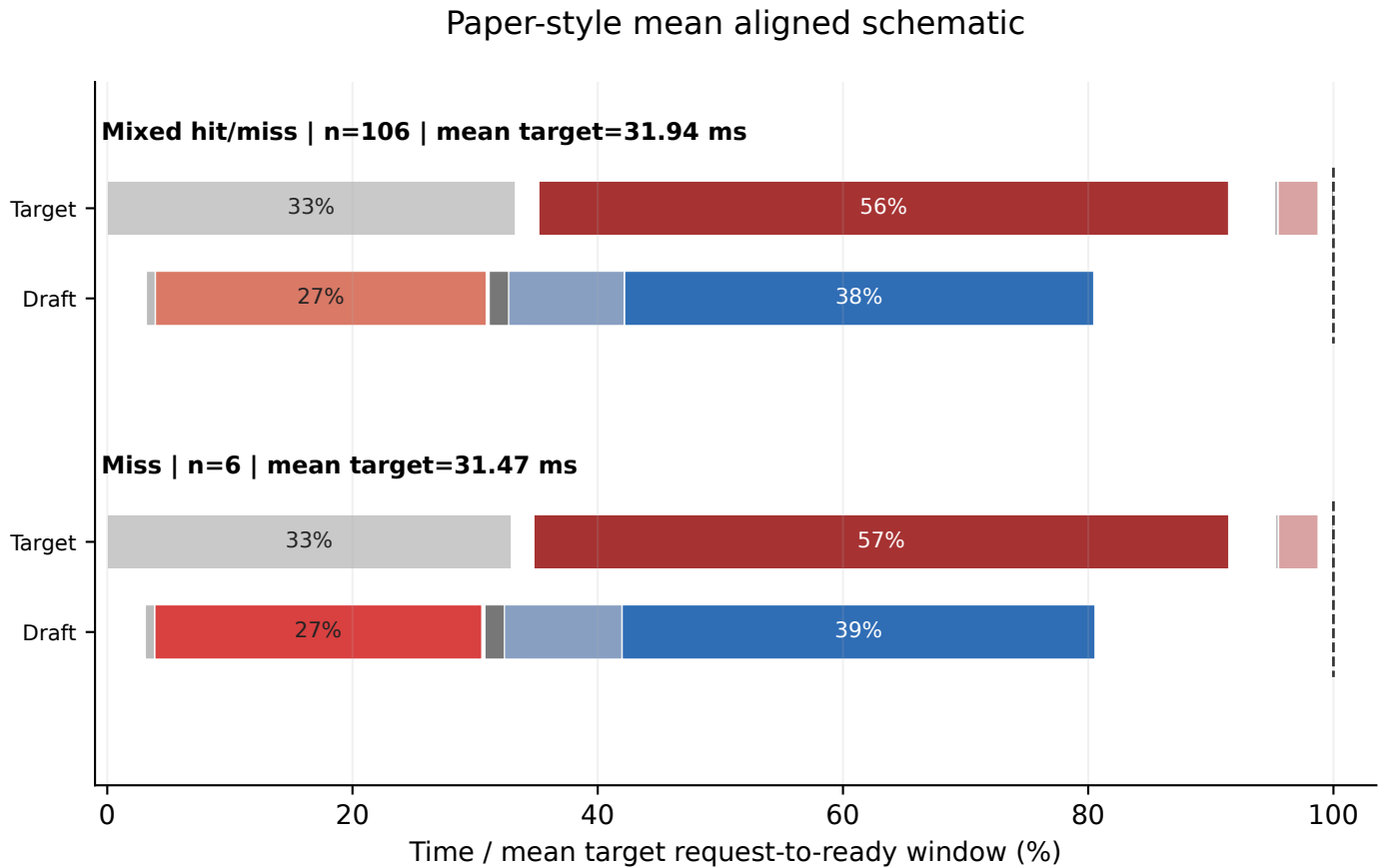
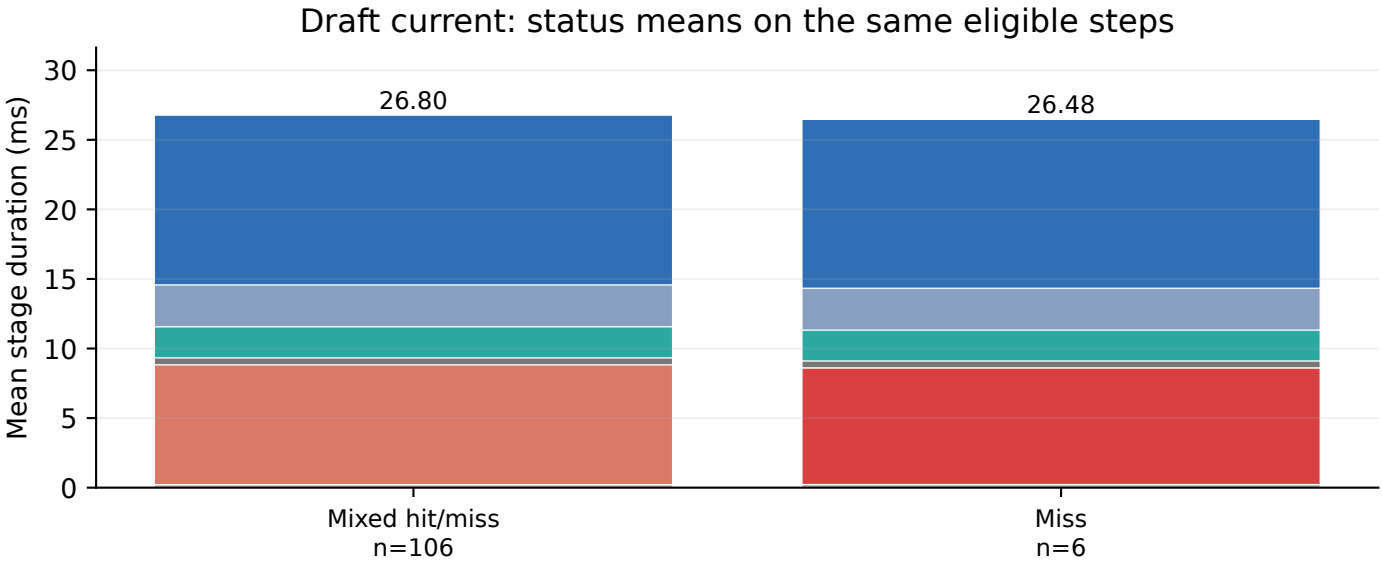
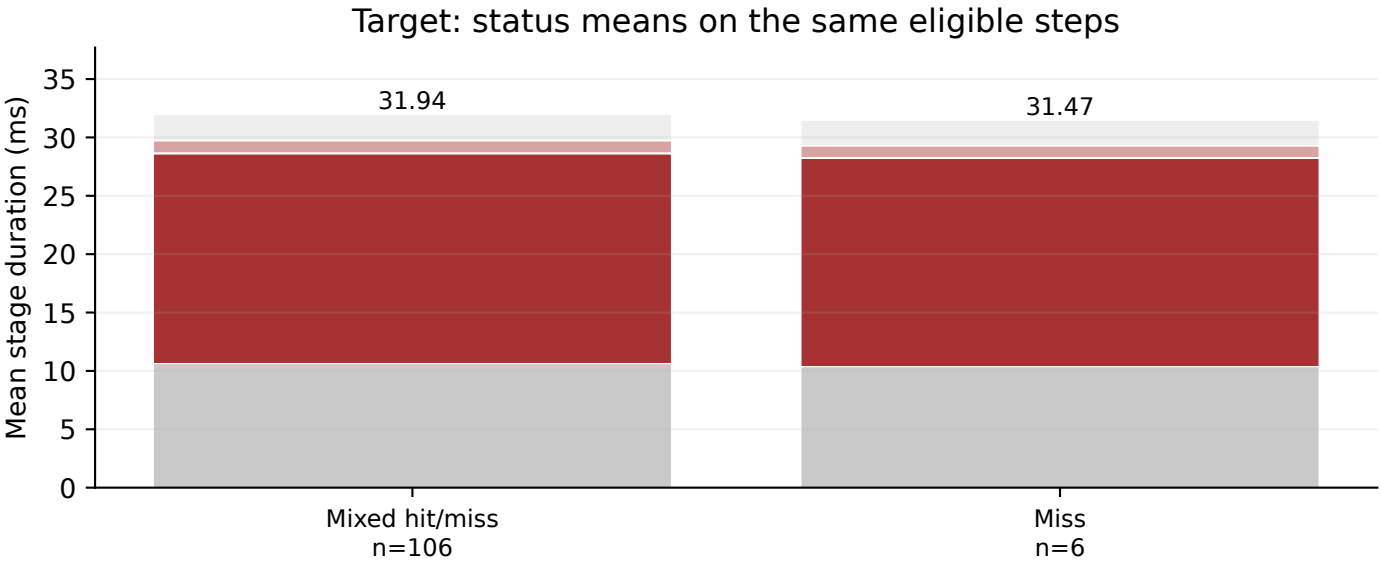


Observed representative steps (near median target window)

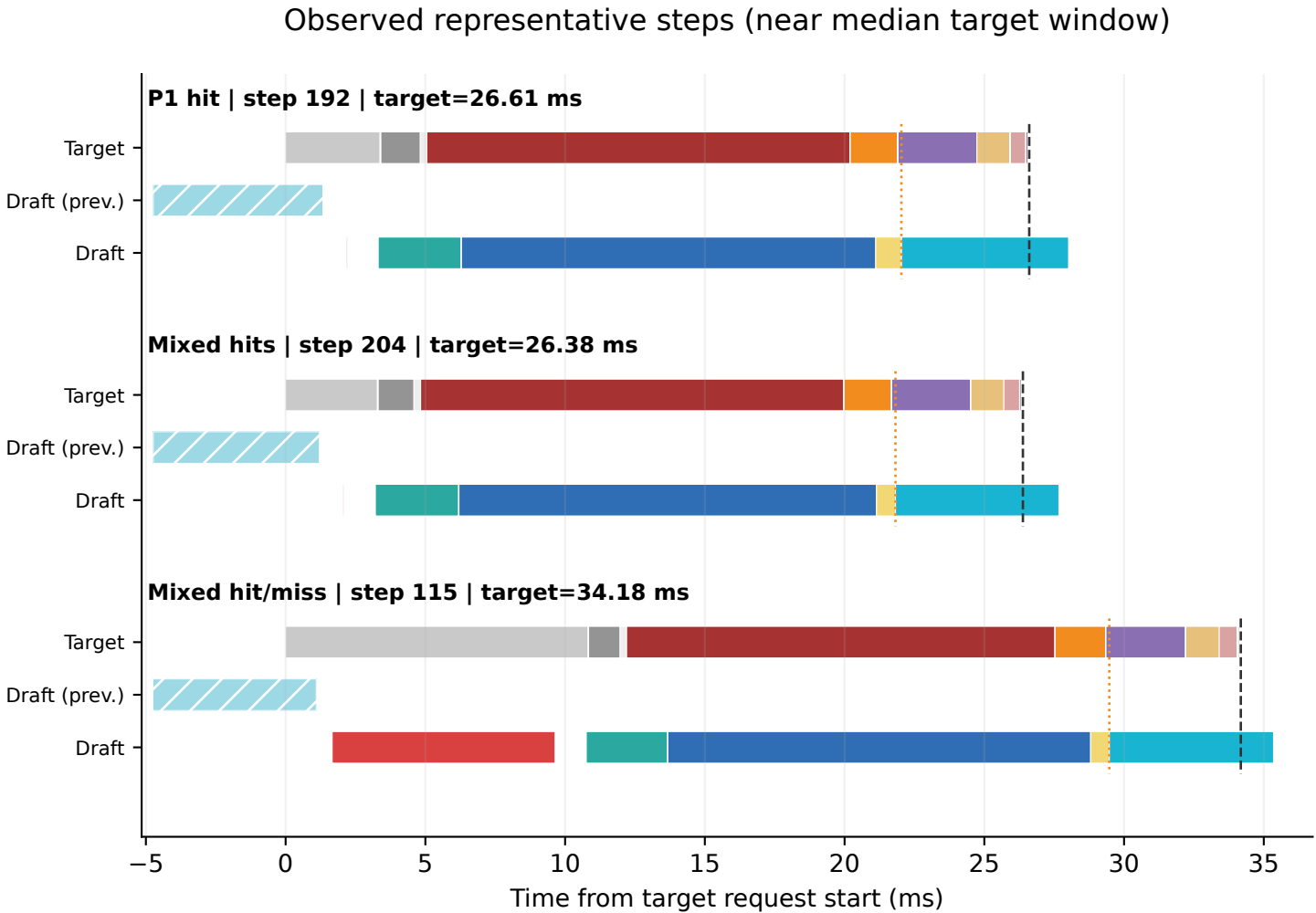
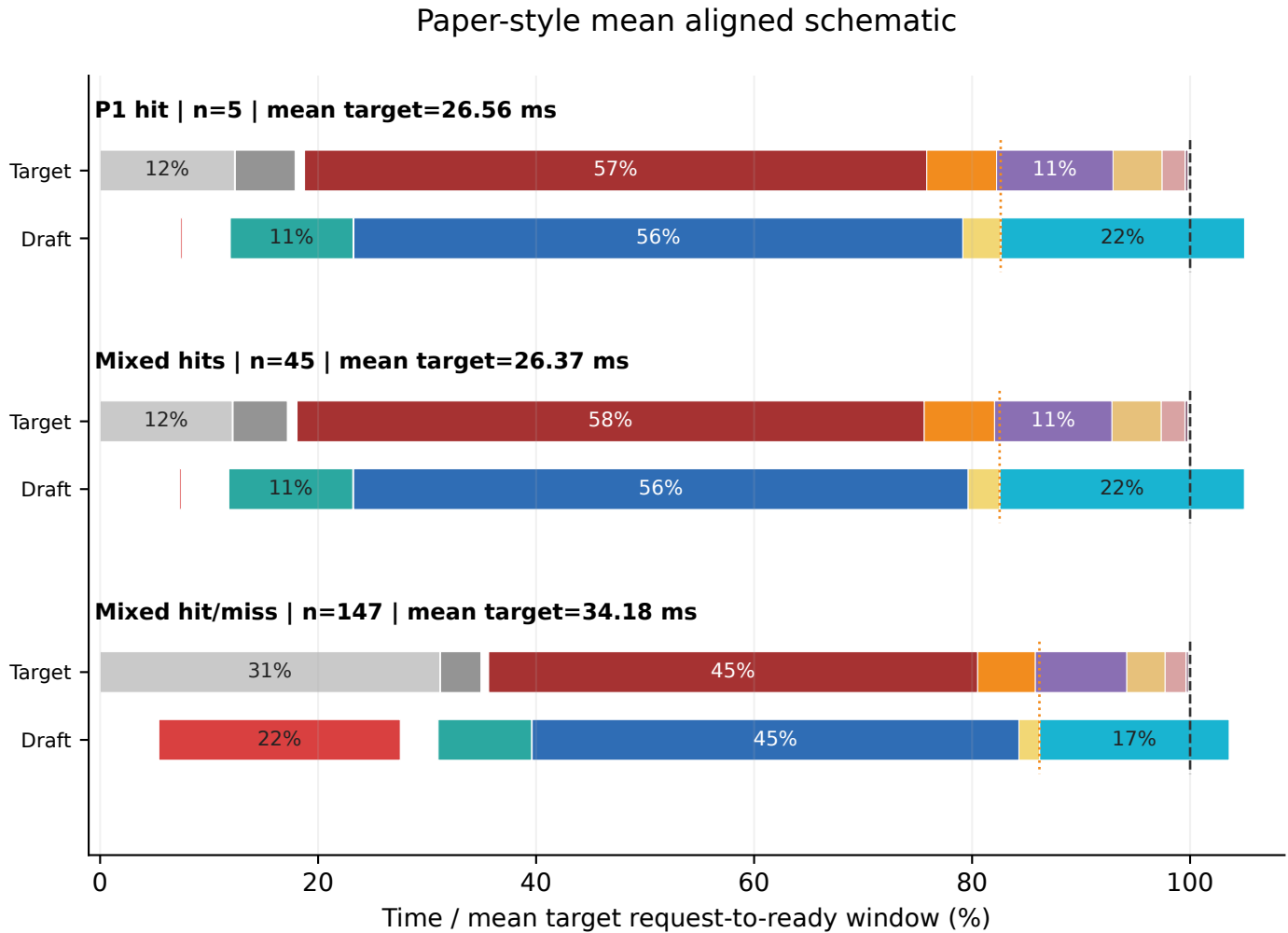
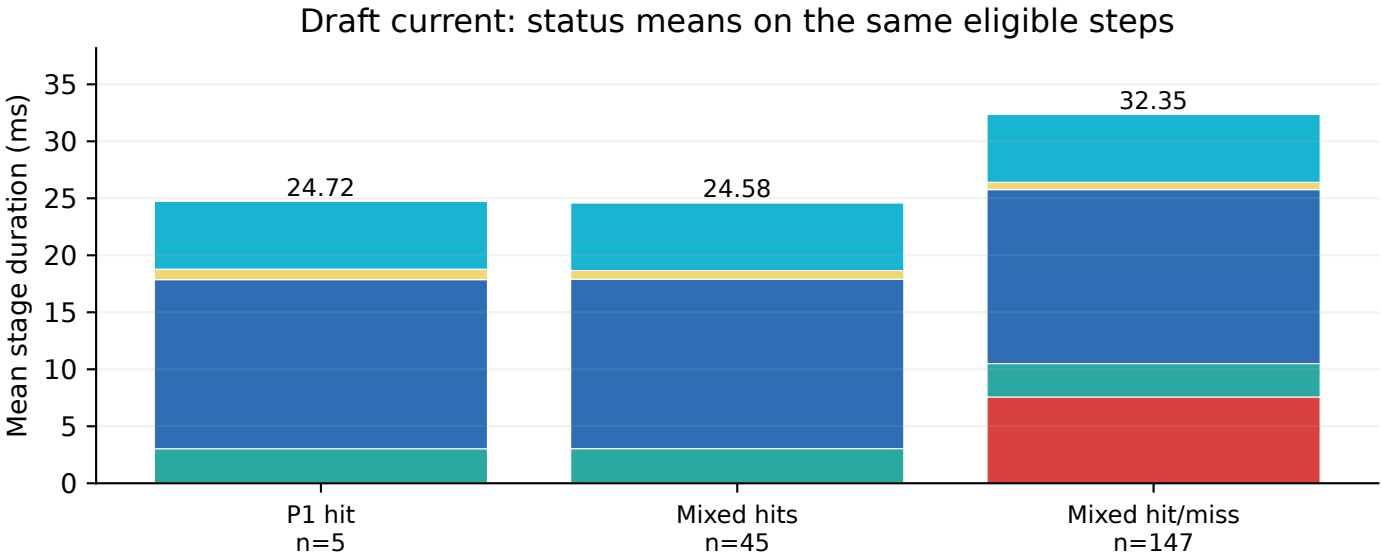
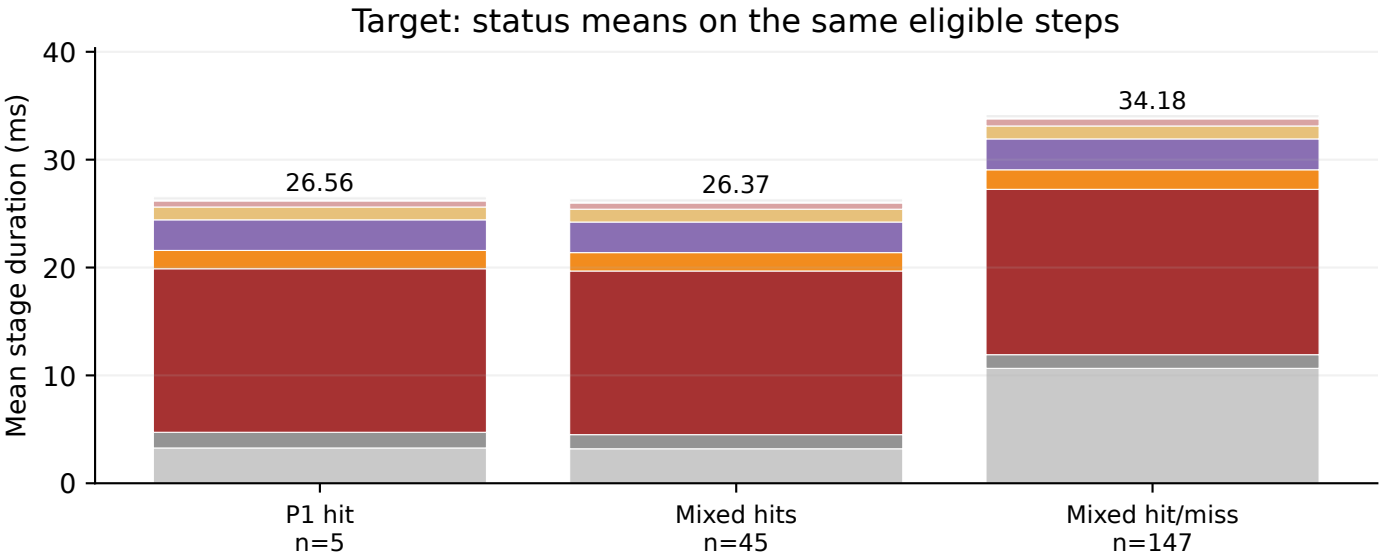


Wait / sync Target post Final logits Postprocess Context P2 Target ready
Prepare Proxy / launch Accept Unlabelled gap P1 Proxy wait Proxy received
Target pre Proxy compute Commit Miss draft

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



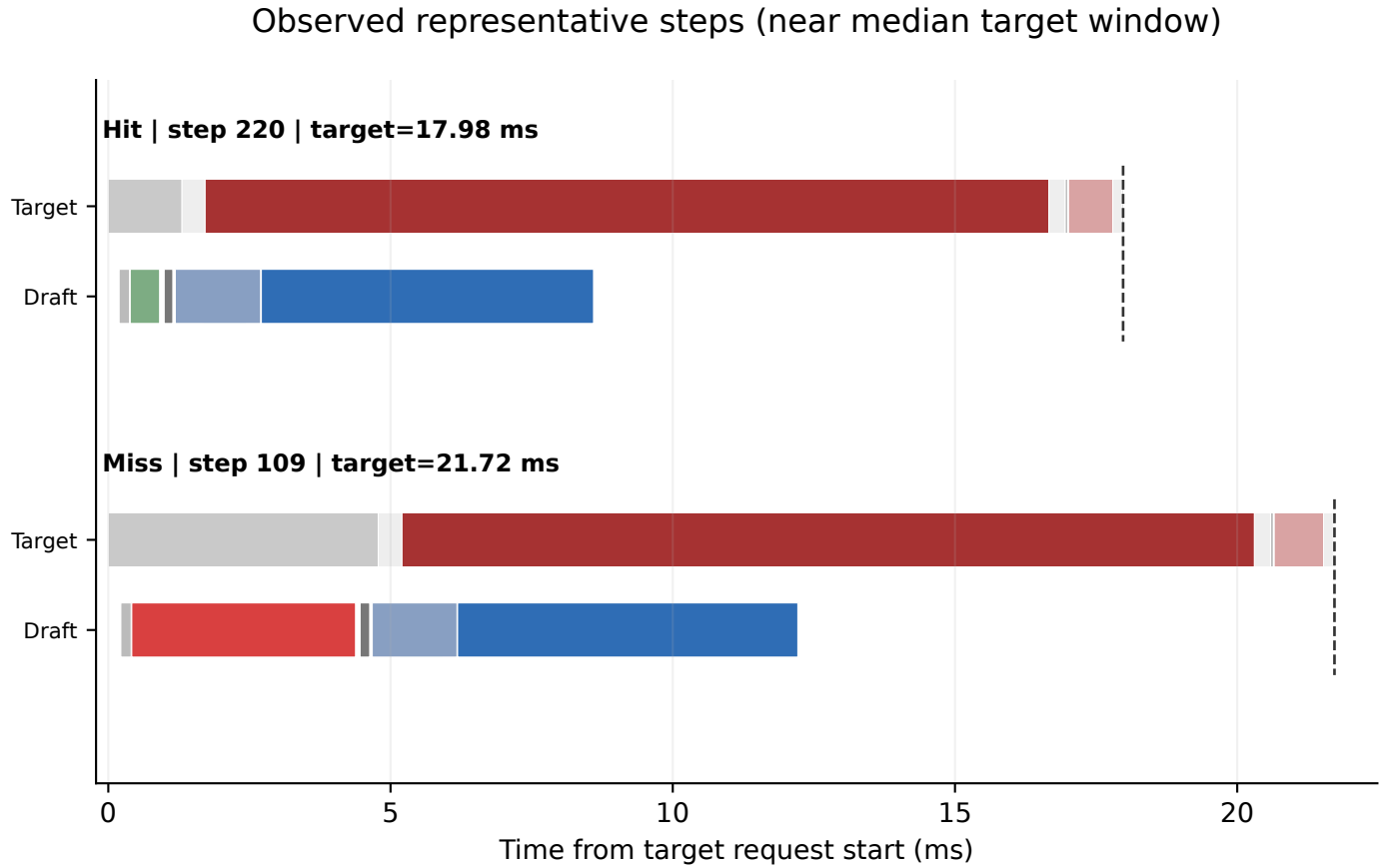
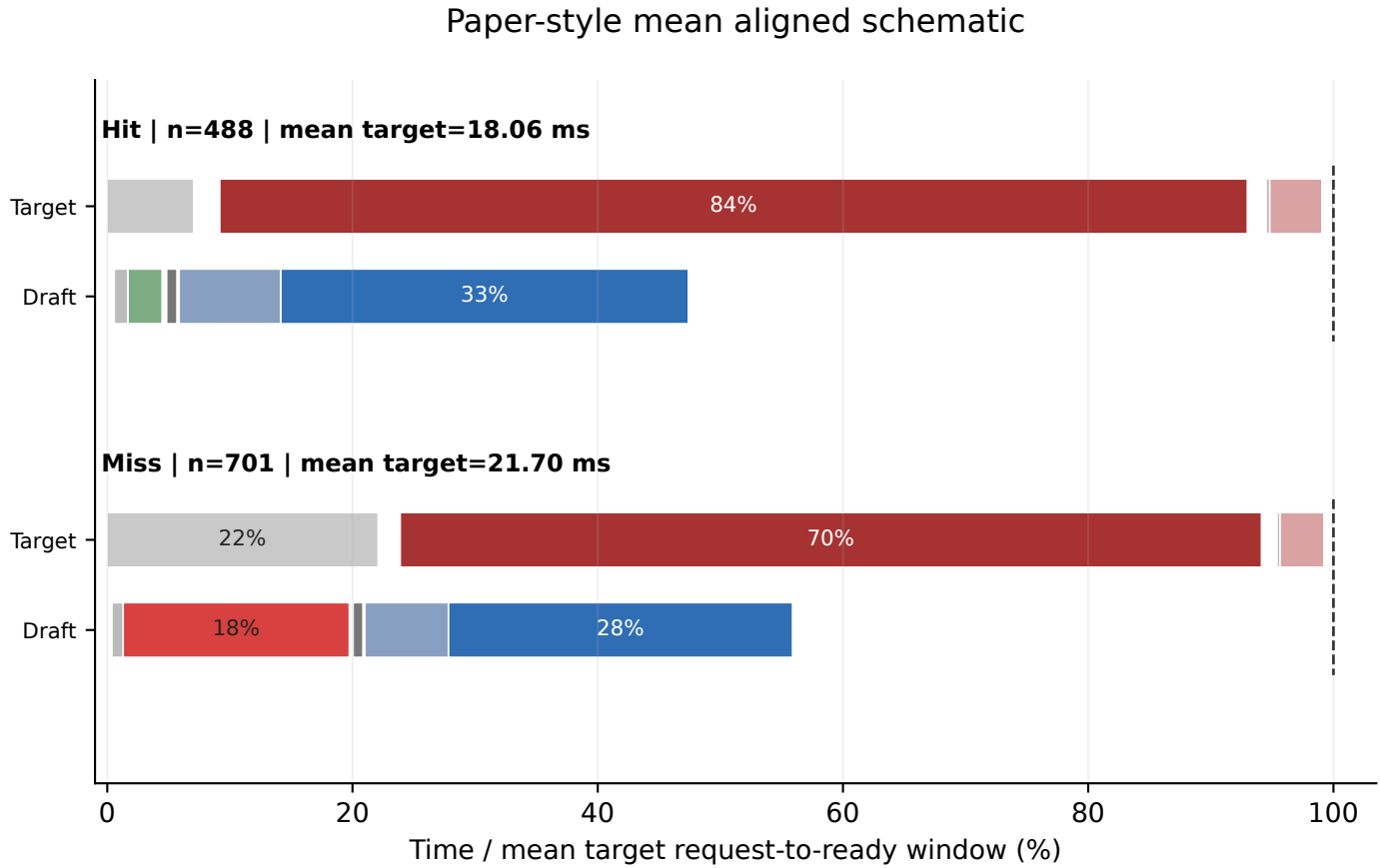
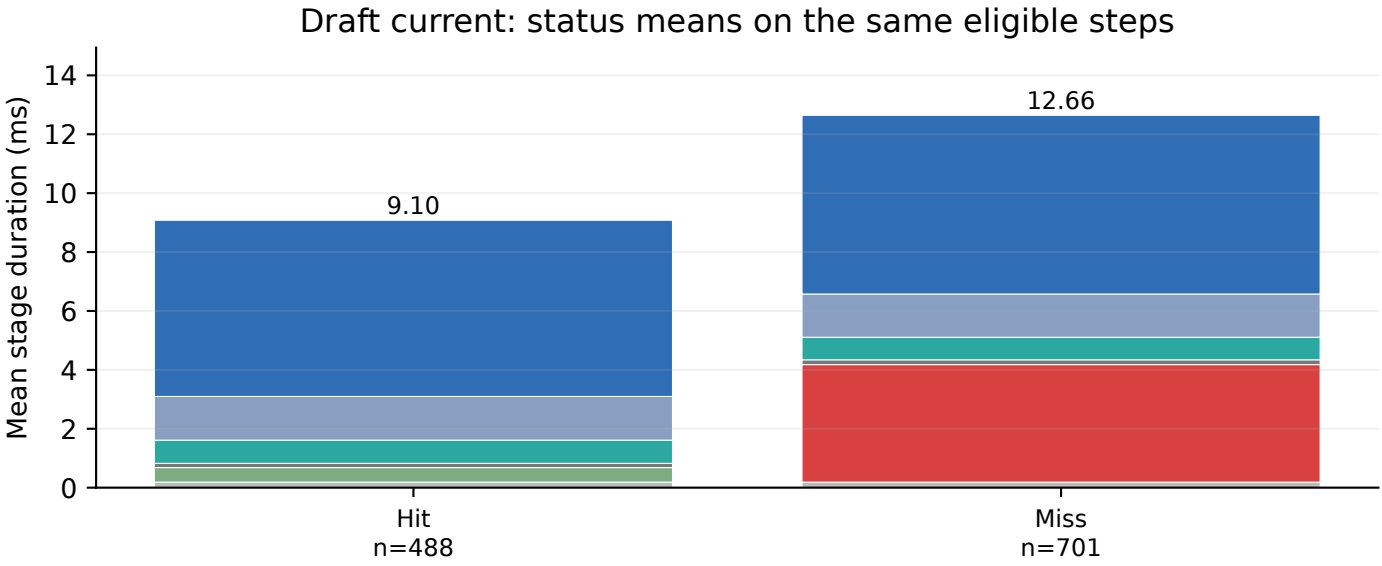
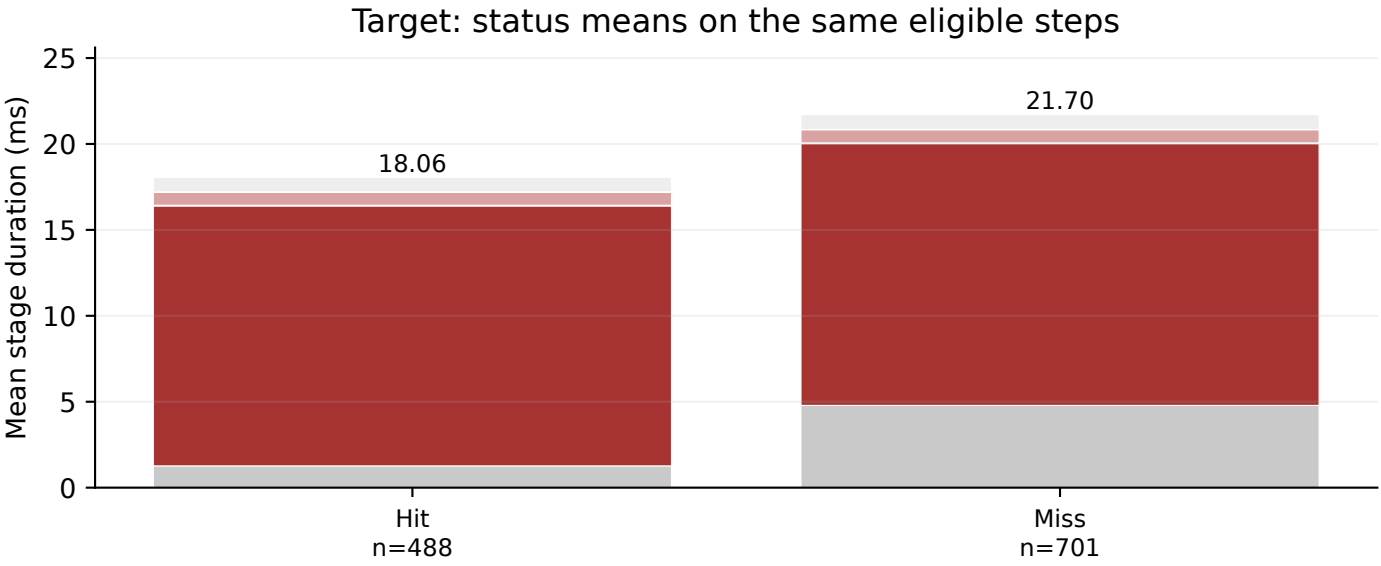
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Target post Accept Unlabelled gap P1 Proxy wait Target ready
Prepare Proxy / launch Commit Miss draft P2 Previous P2 Proxy received

Target pre Final logits Postprocess Context

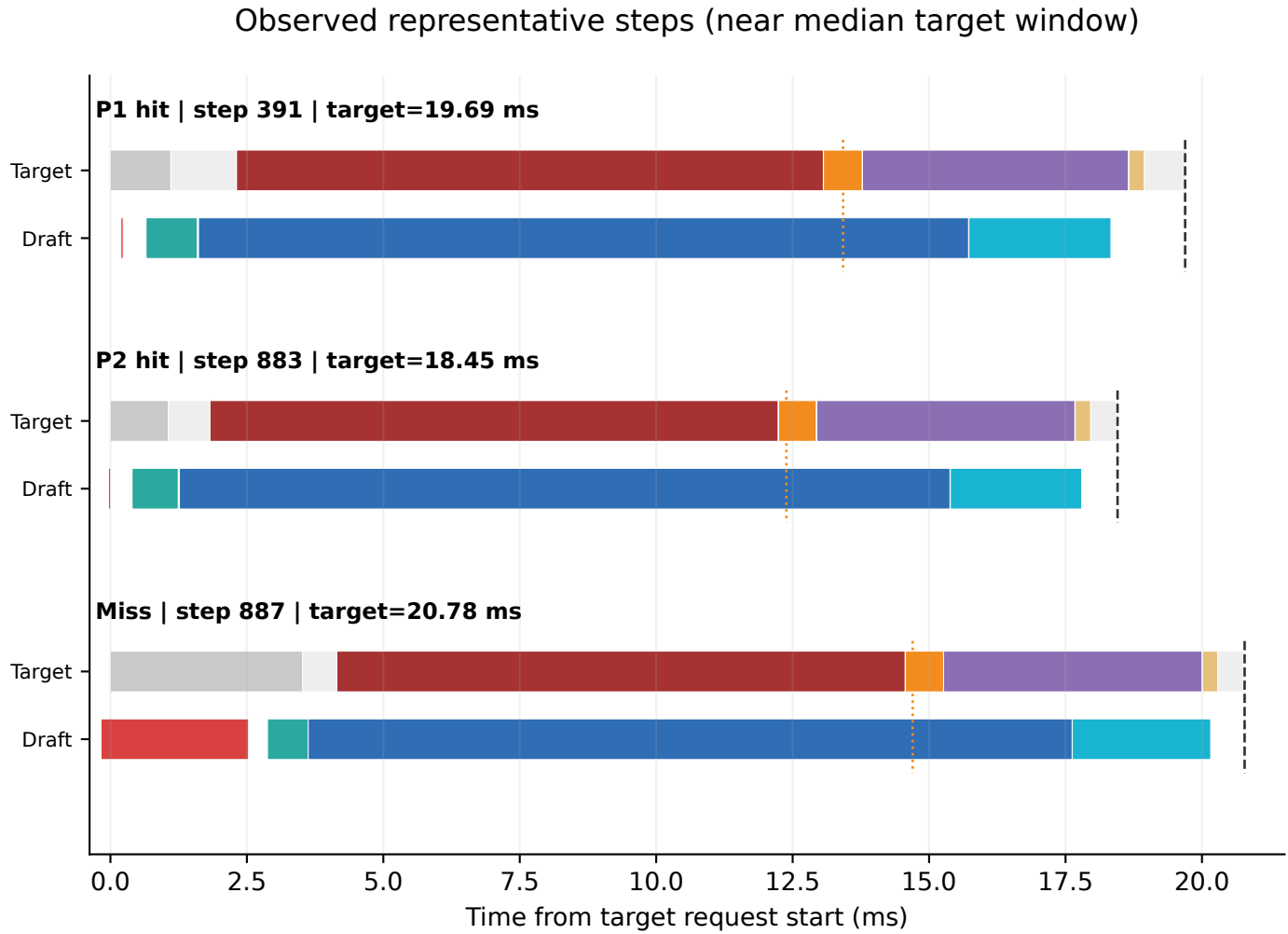
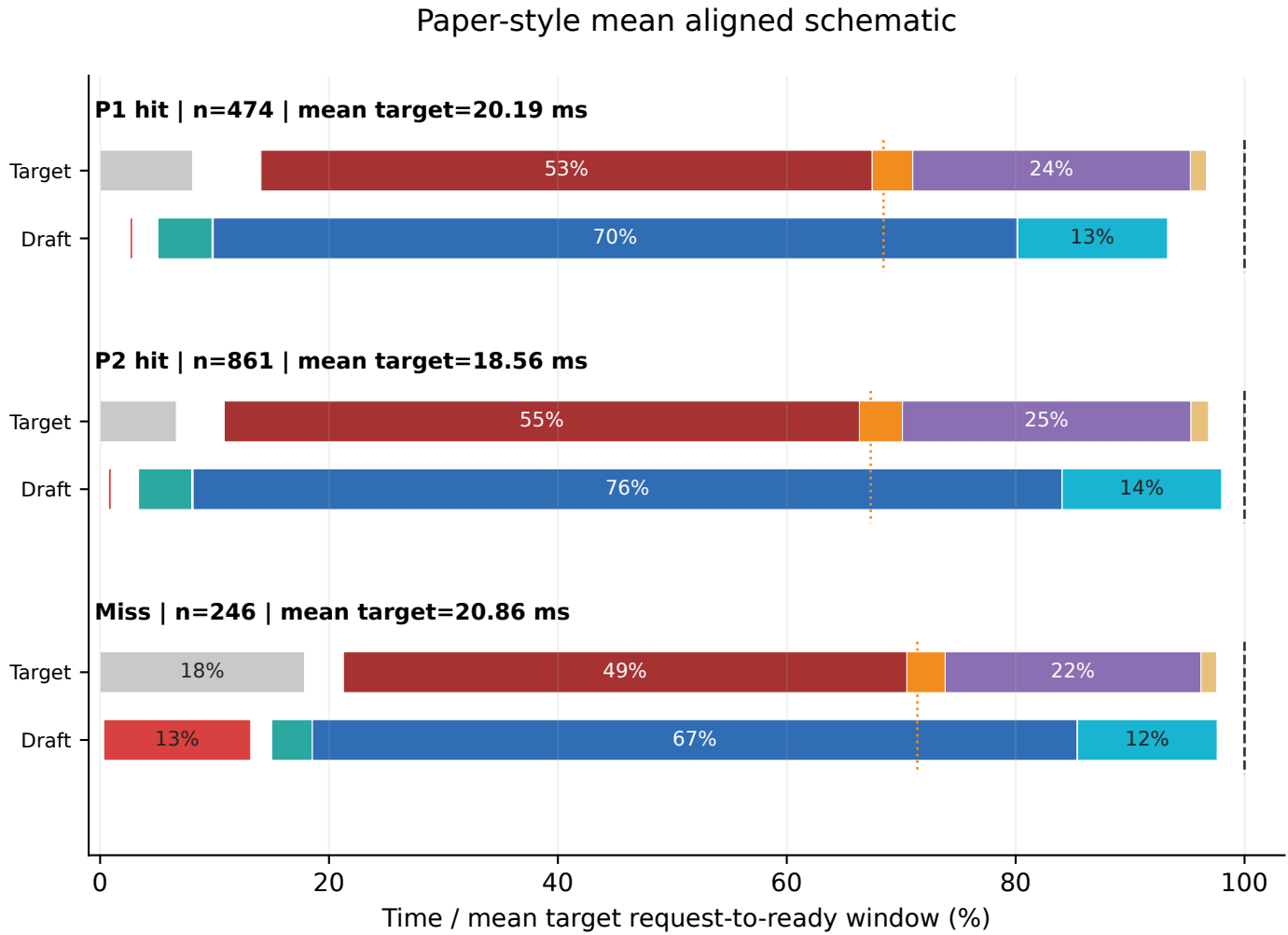
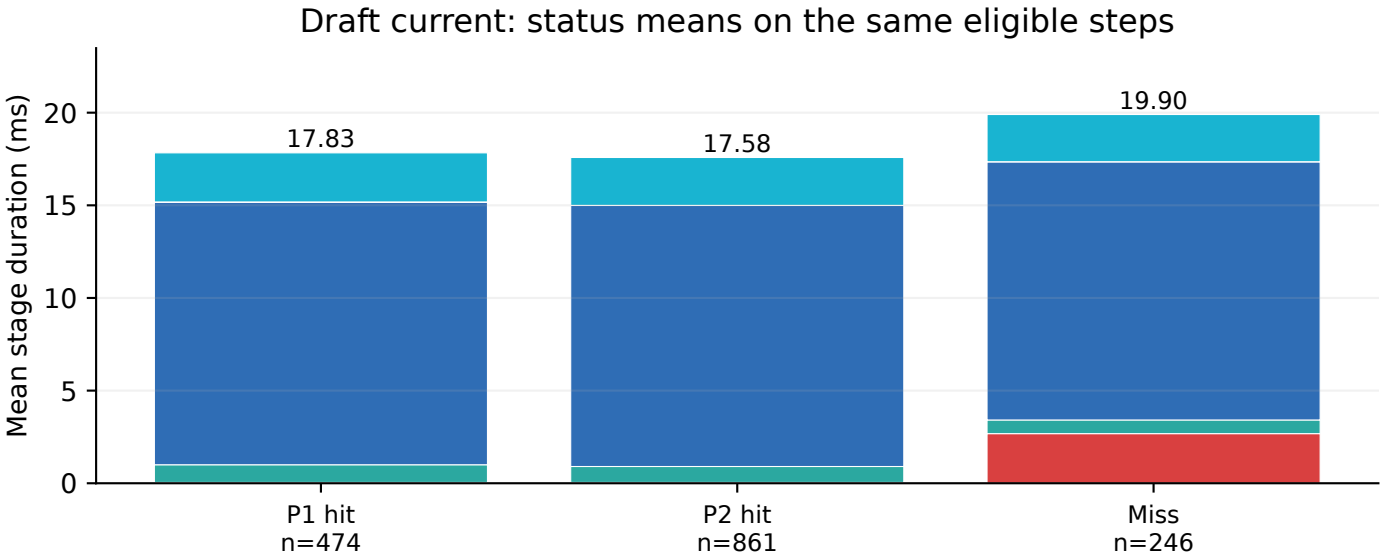
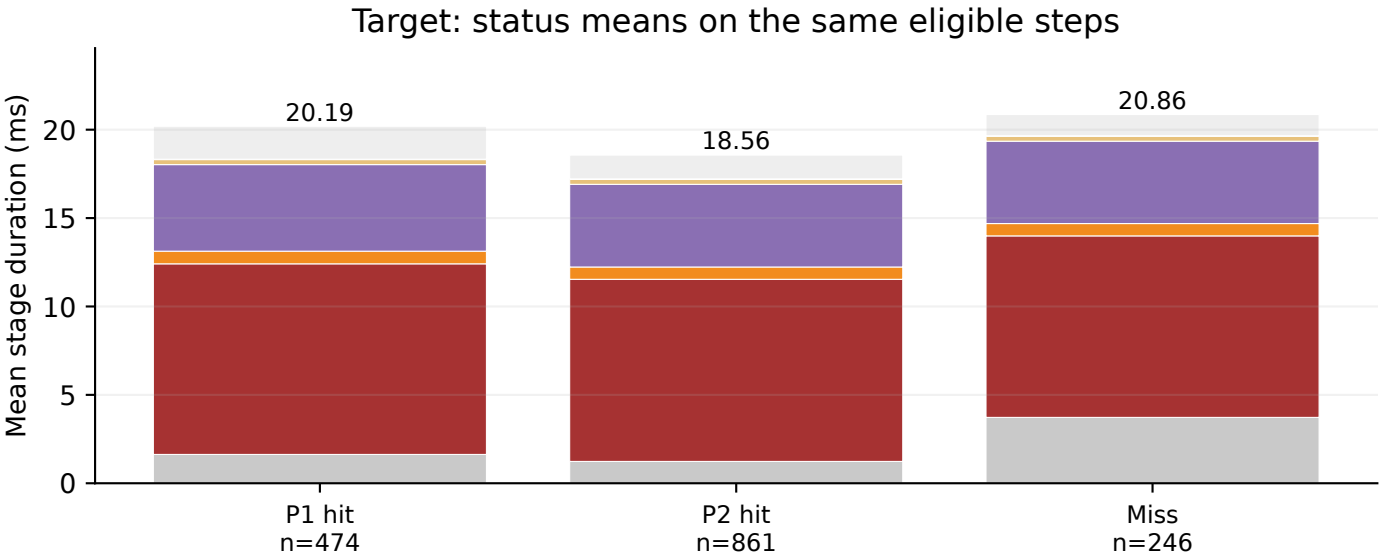
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



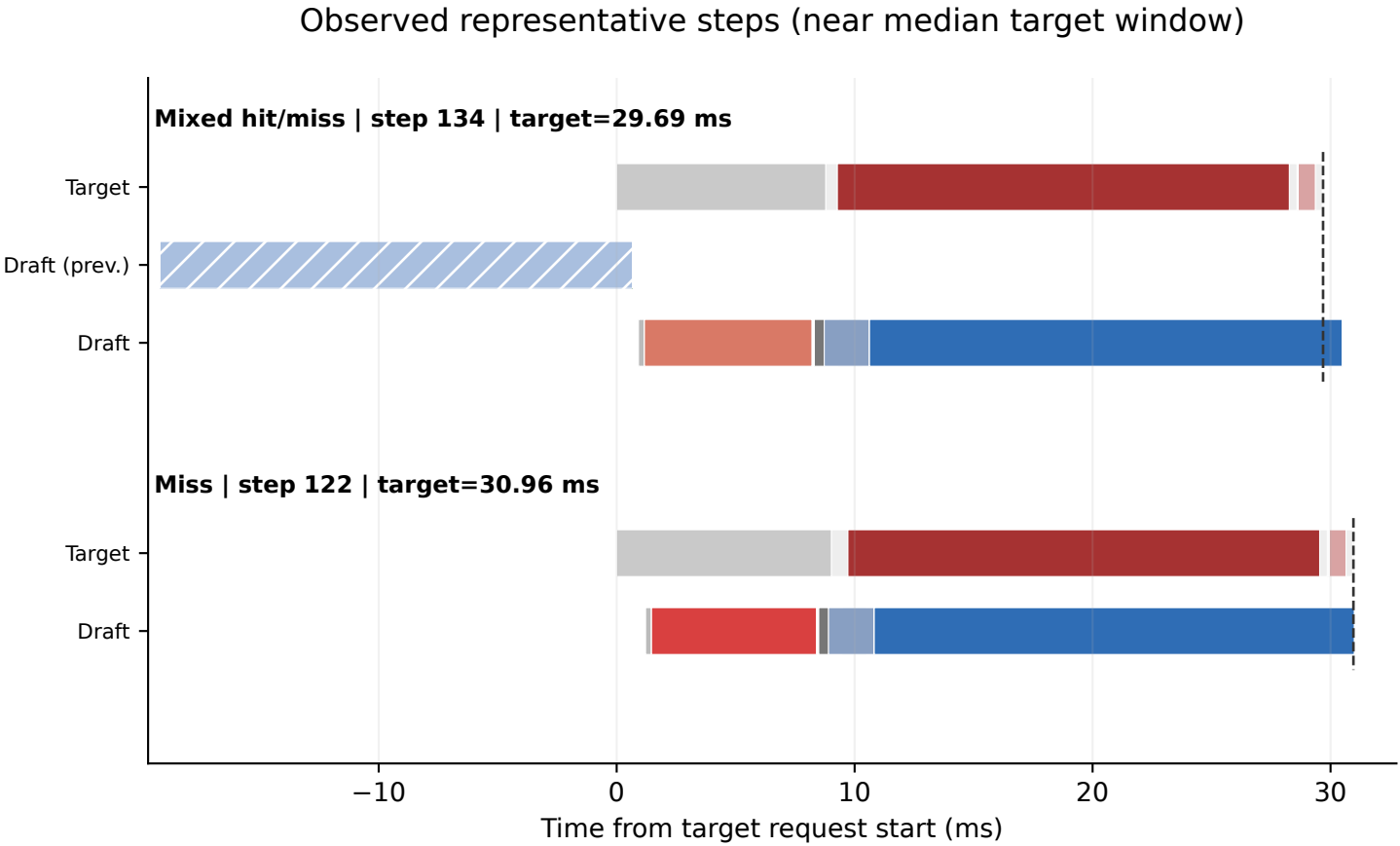
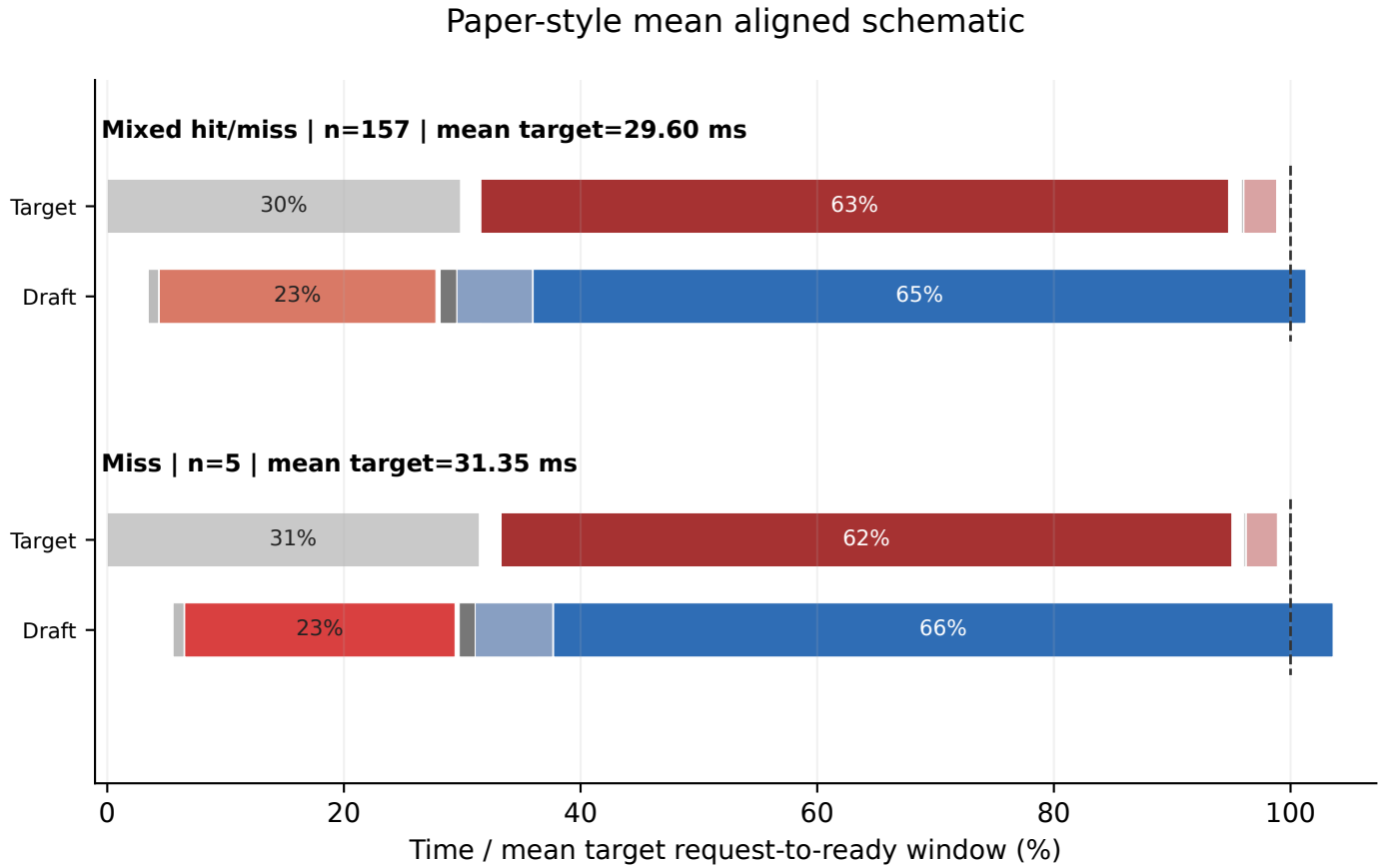
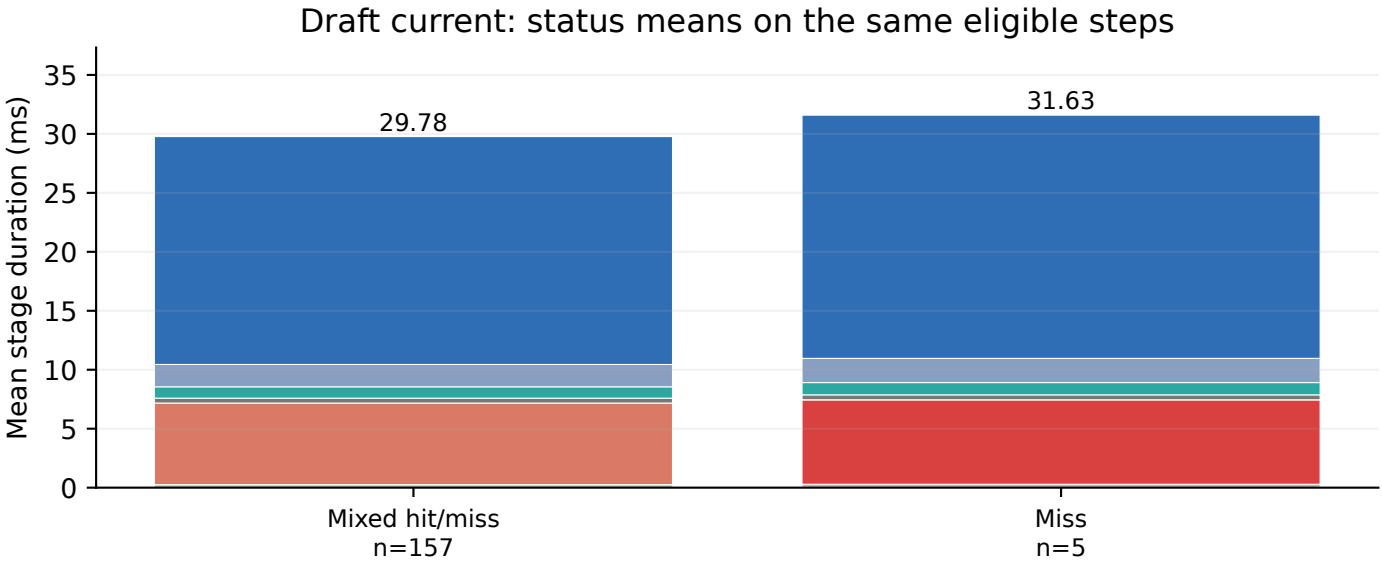
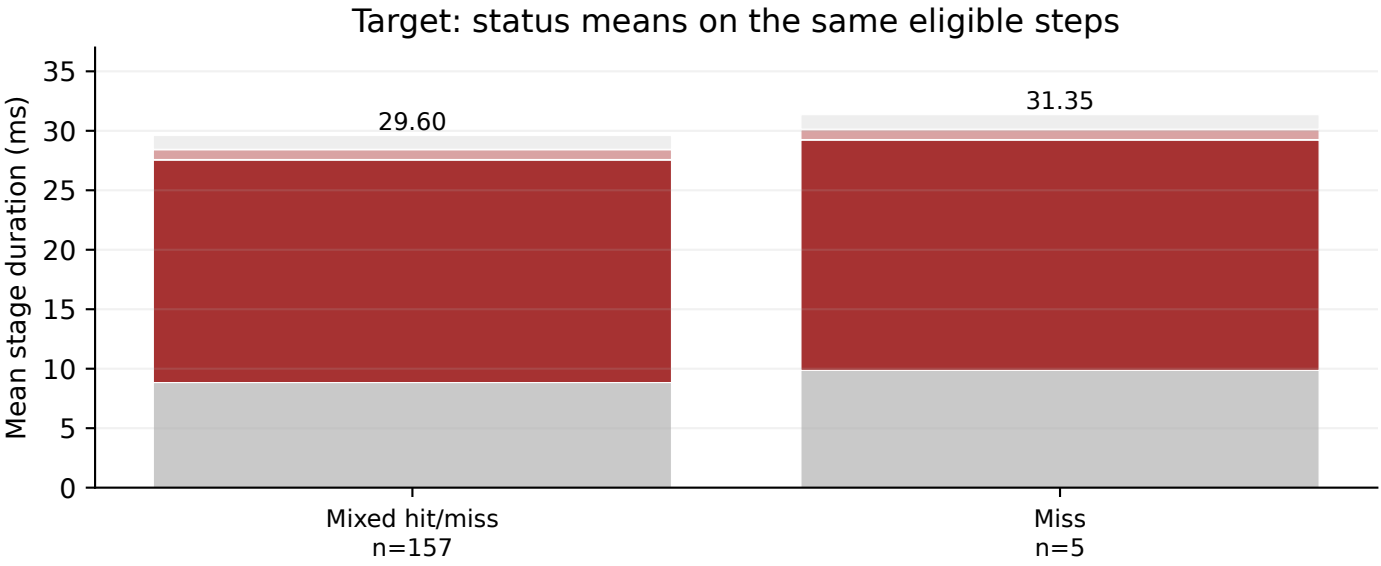
Wait / sync Accept prepare Miss response Request Send response SSD decode Target ready
Target verify Postprocess Context Cache response SSD build Cache insert Proxy received
Accept Unlabelled gap

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b1_refine_e21_k12_2_r2
DUET: K1/K2=12/2, exit=21 (0-based), C=3, N1/N2=16/4, U=2, W=13, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1581 eligible steps



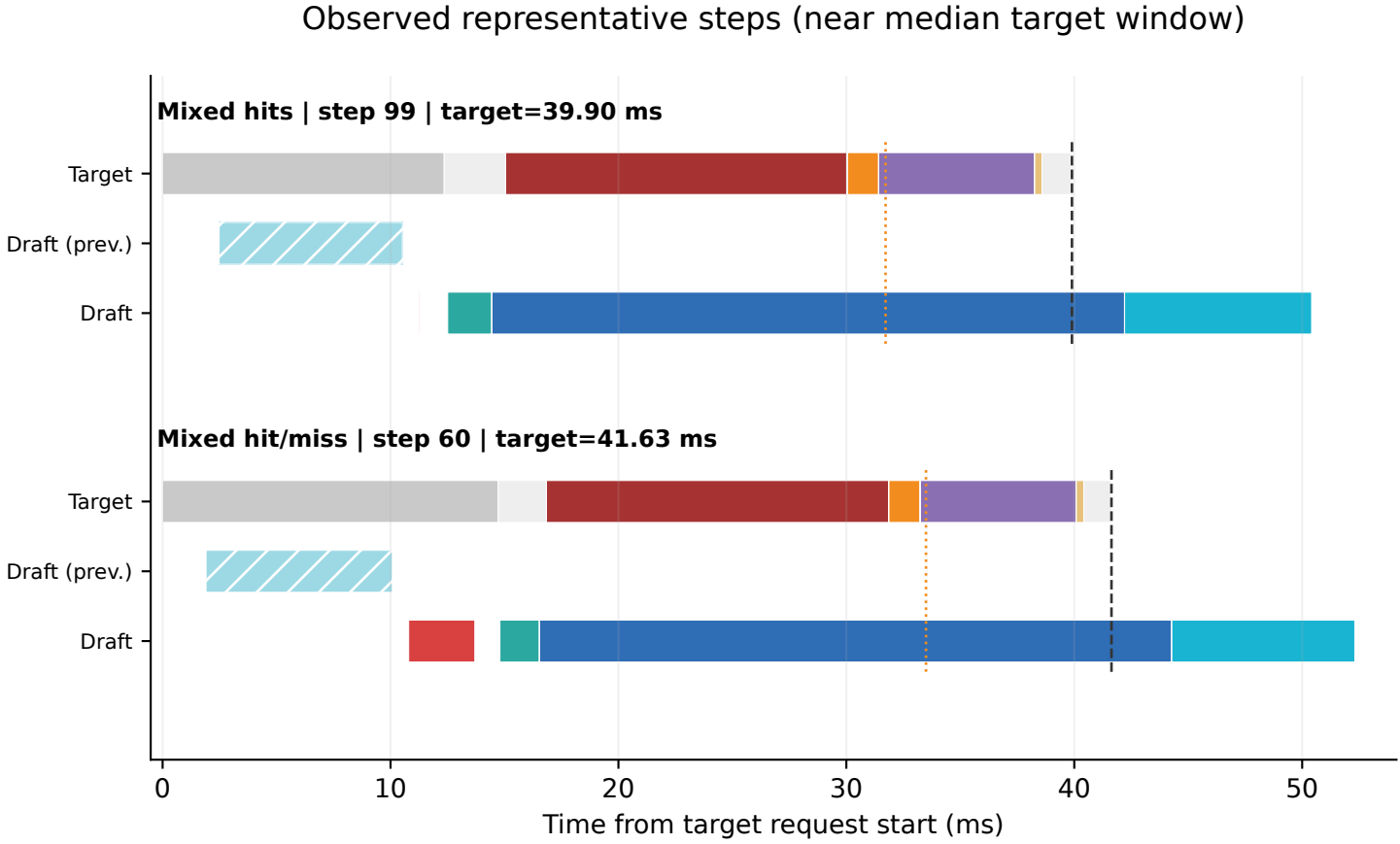
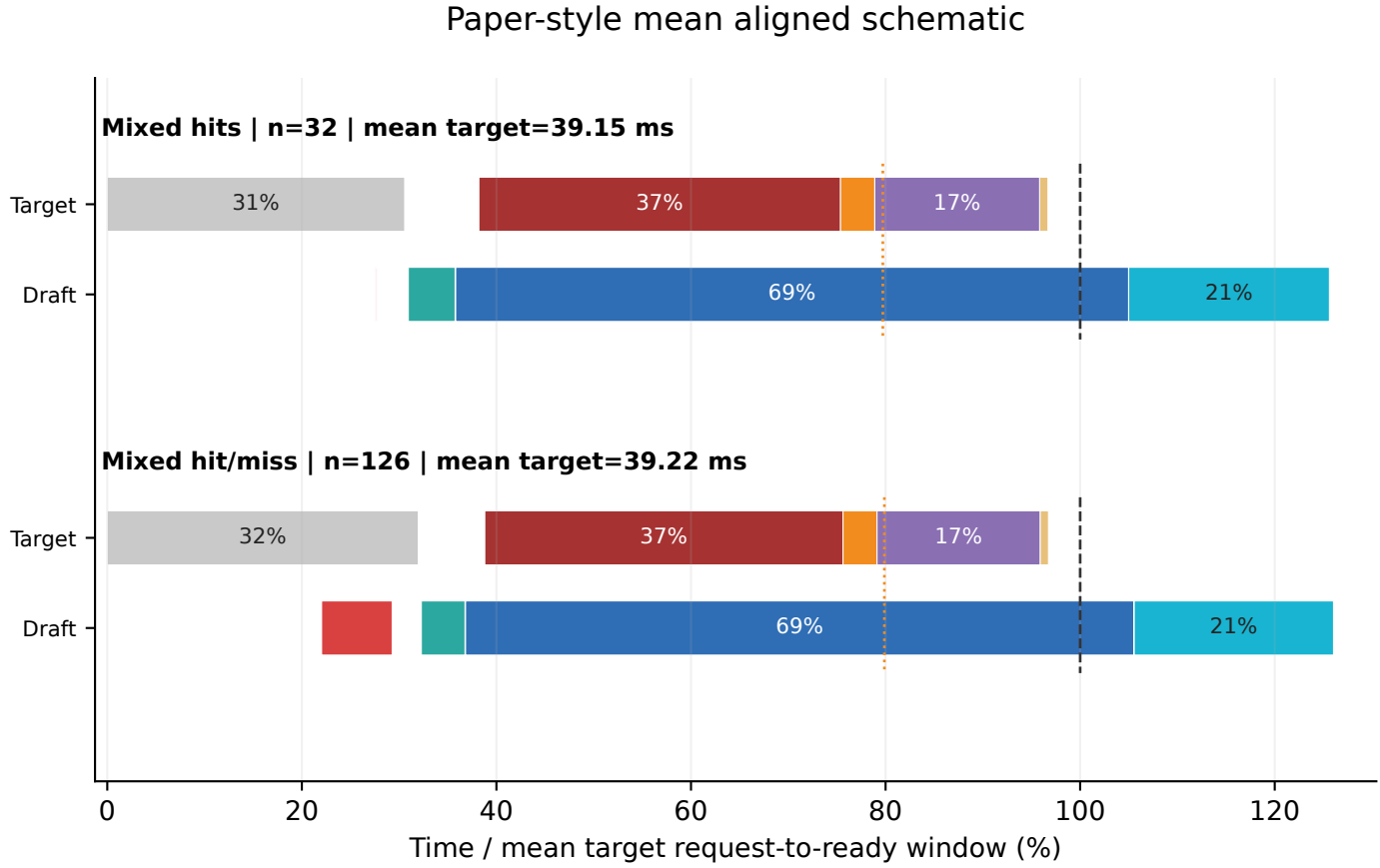
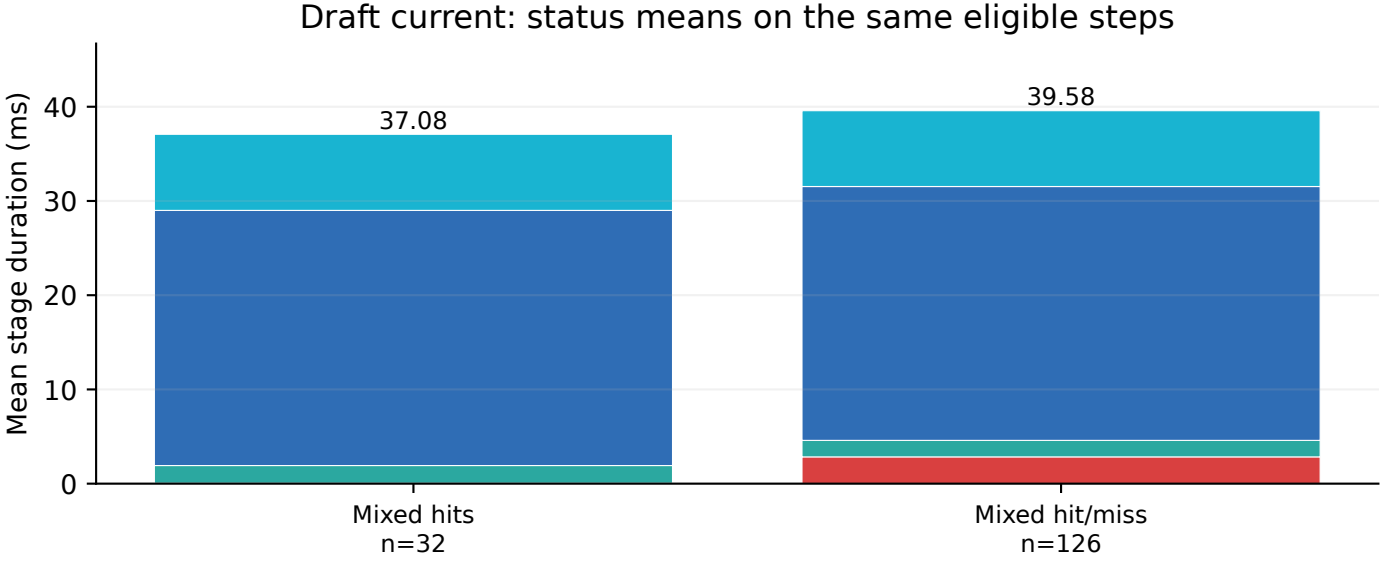
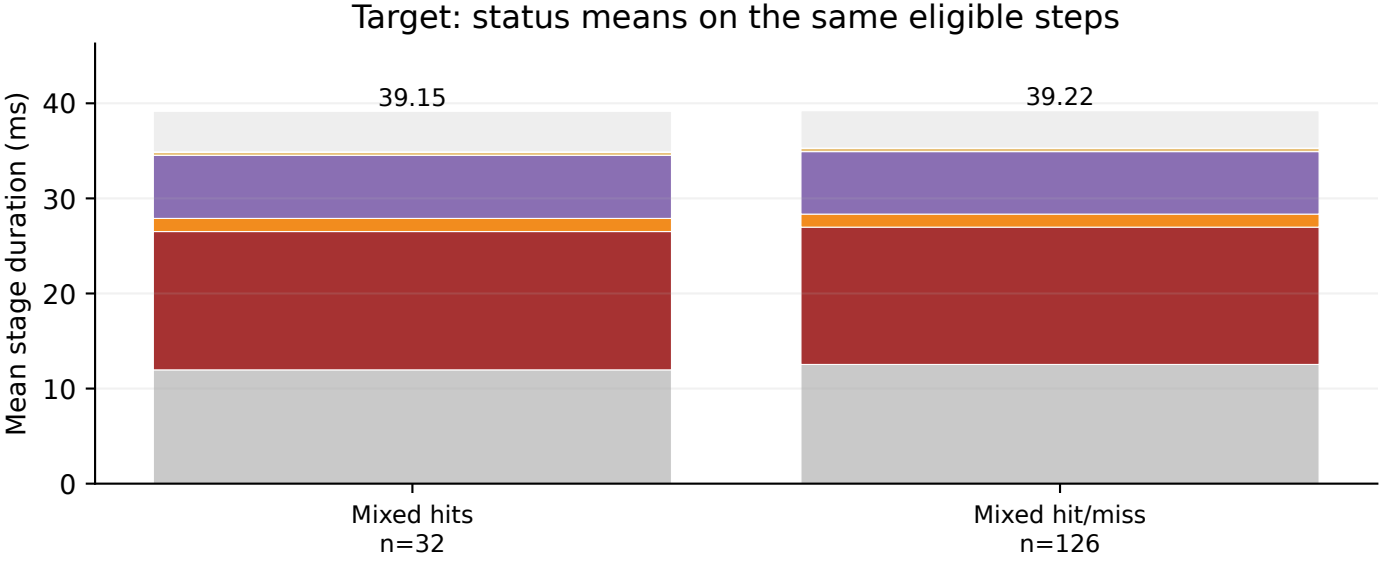
Wait / sync Target post Final logits Unlabelled gap Context P2 Target ready
Target pre Proxy / launch Postprocess Miss draft P1 Proxy wait Proxy received
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Miss response Request SSD decode Previous SSD decode Target ready
Target verify Postprocess Mixed response Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Context SSD build

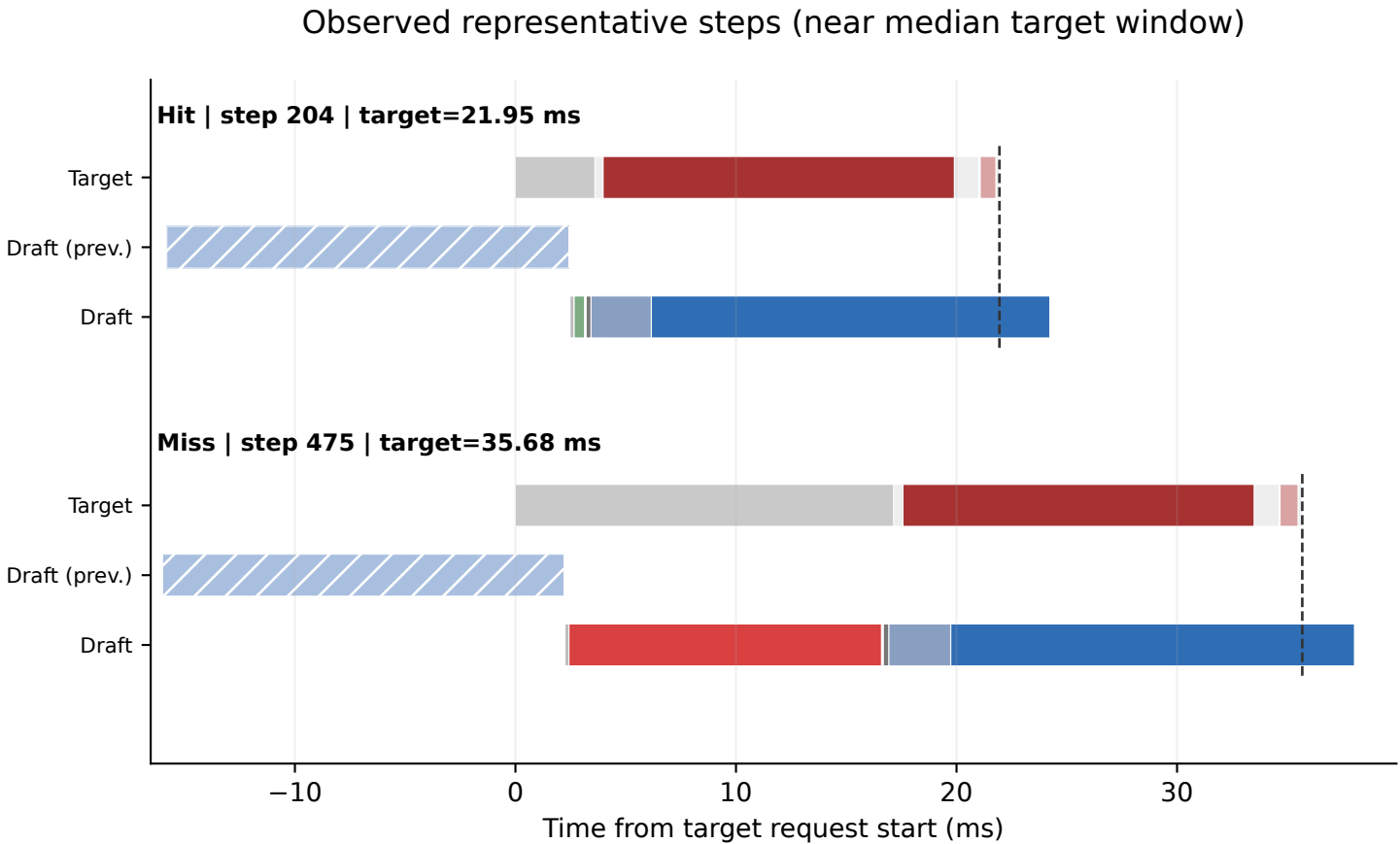
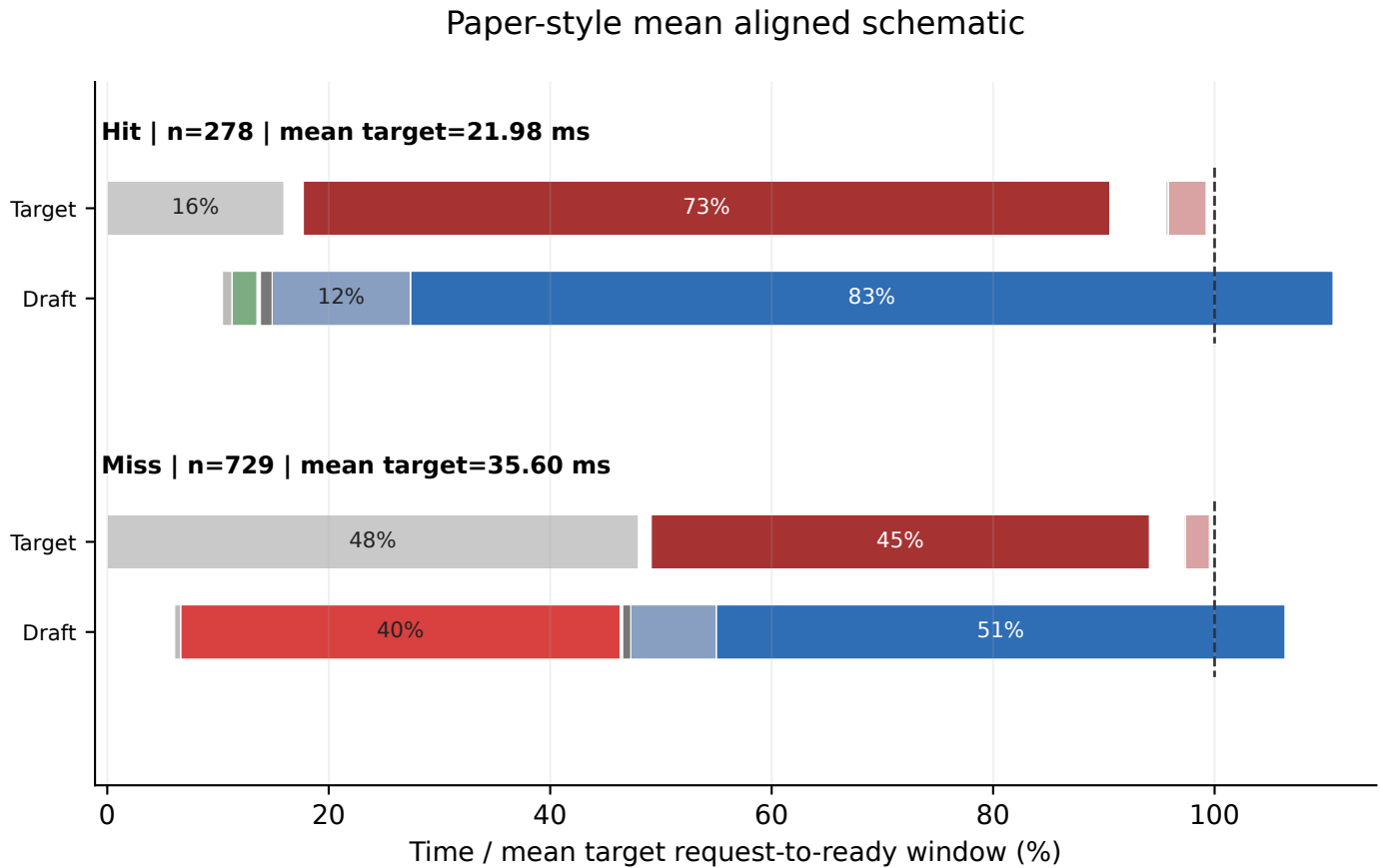
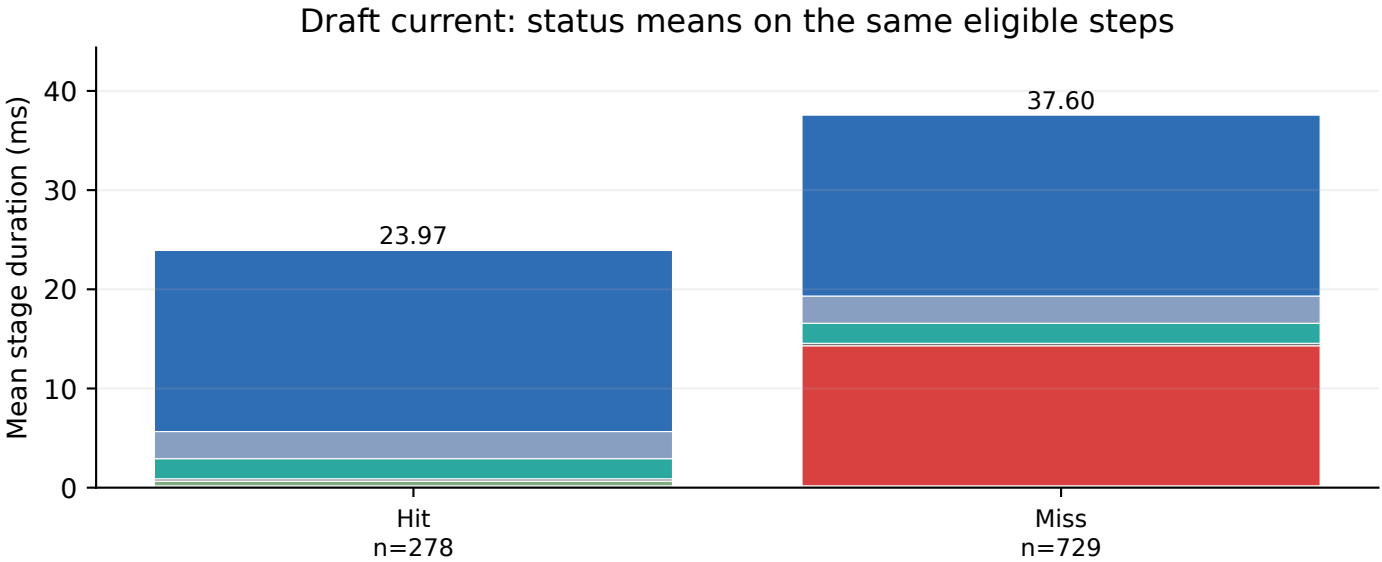
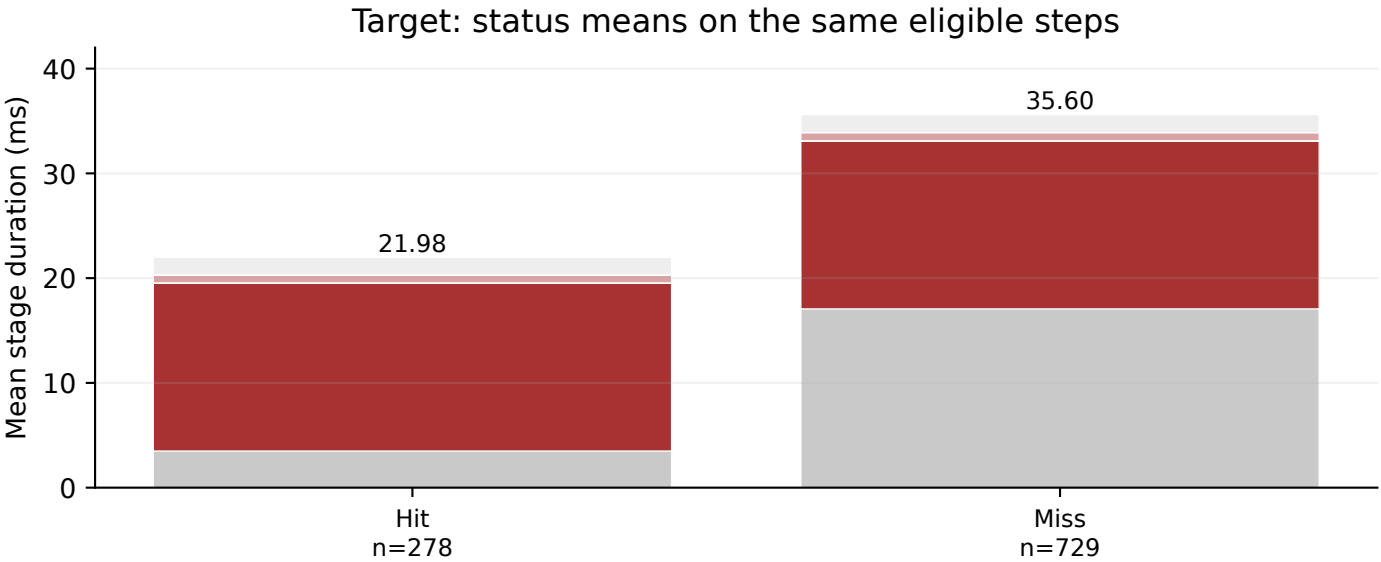
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama2_b8_tree_al_compute3
DUET: K1/K2=6/2, exit=21 (0-based), $\bar{C}$ =3, $\bar{N1}/\bar{N2}$ =12/4, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 158 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

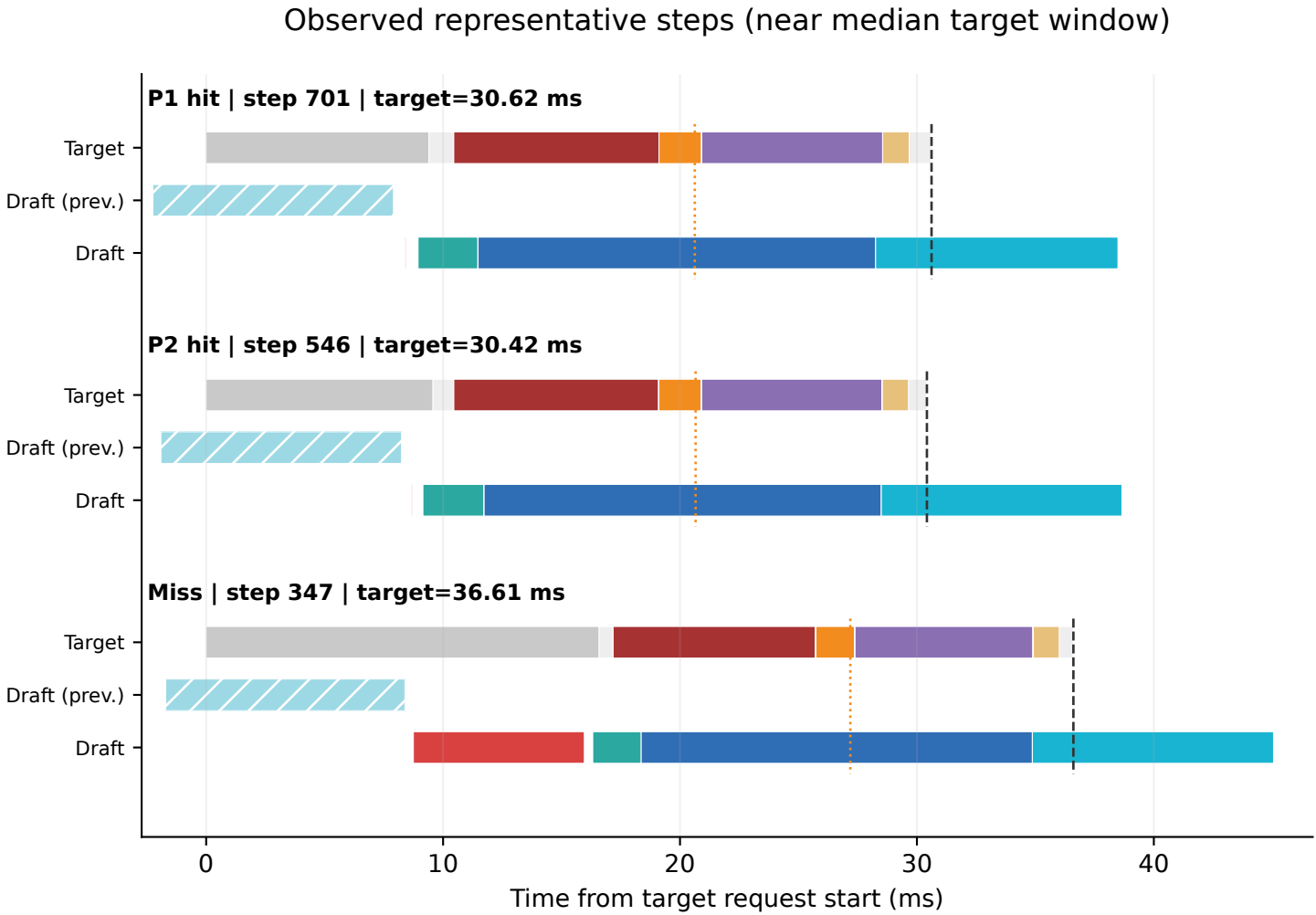
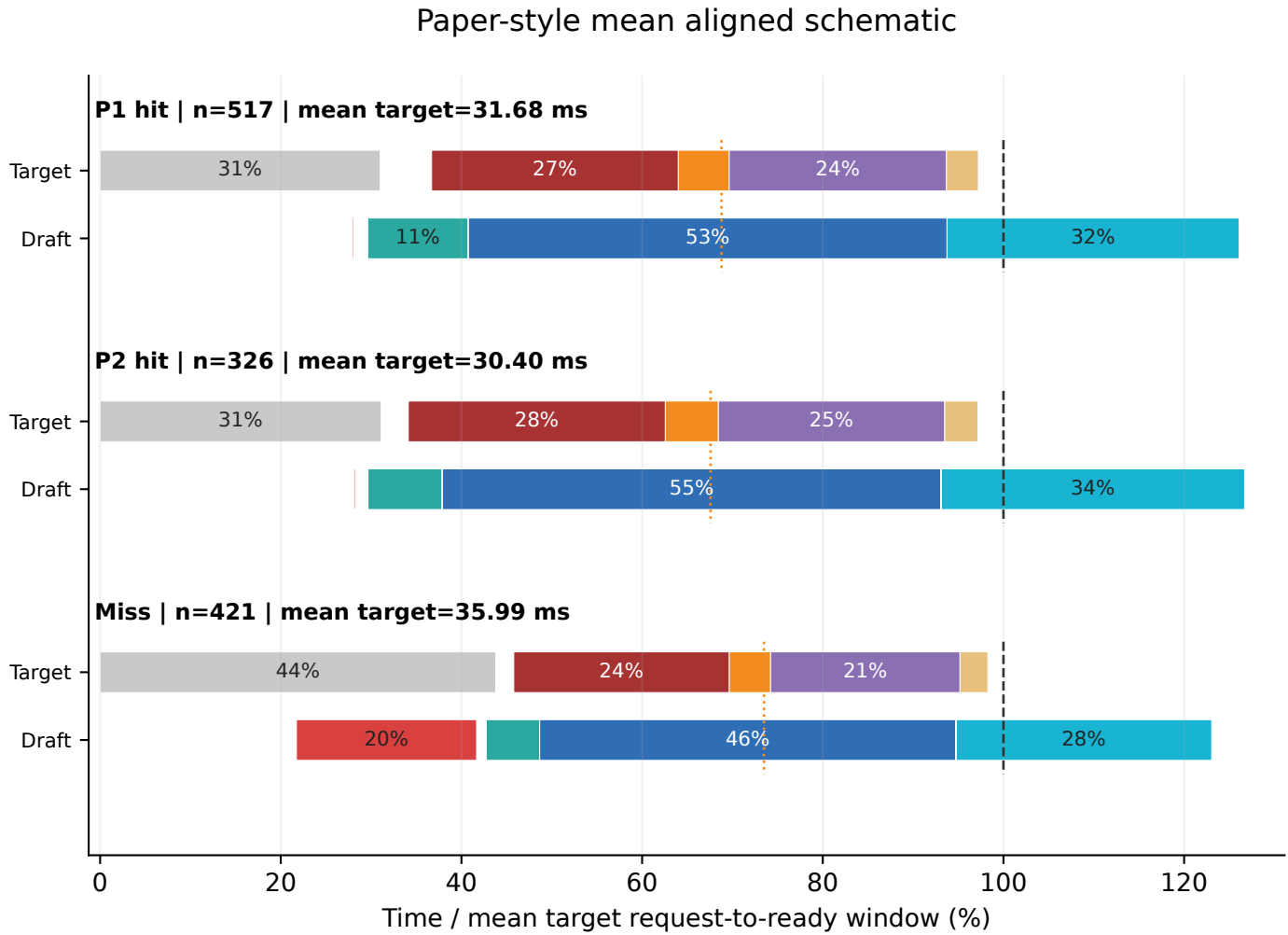
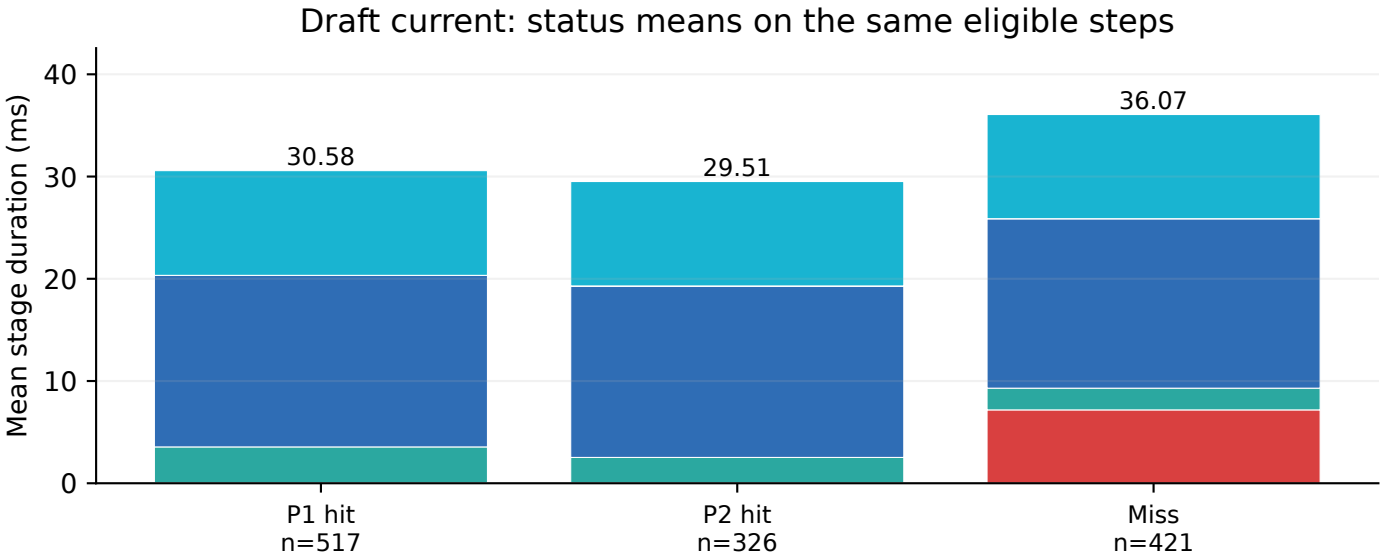
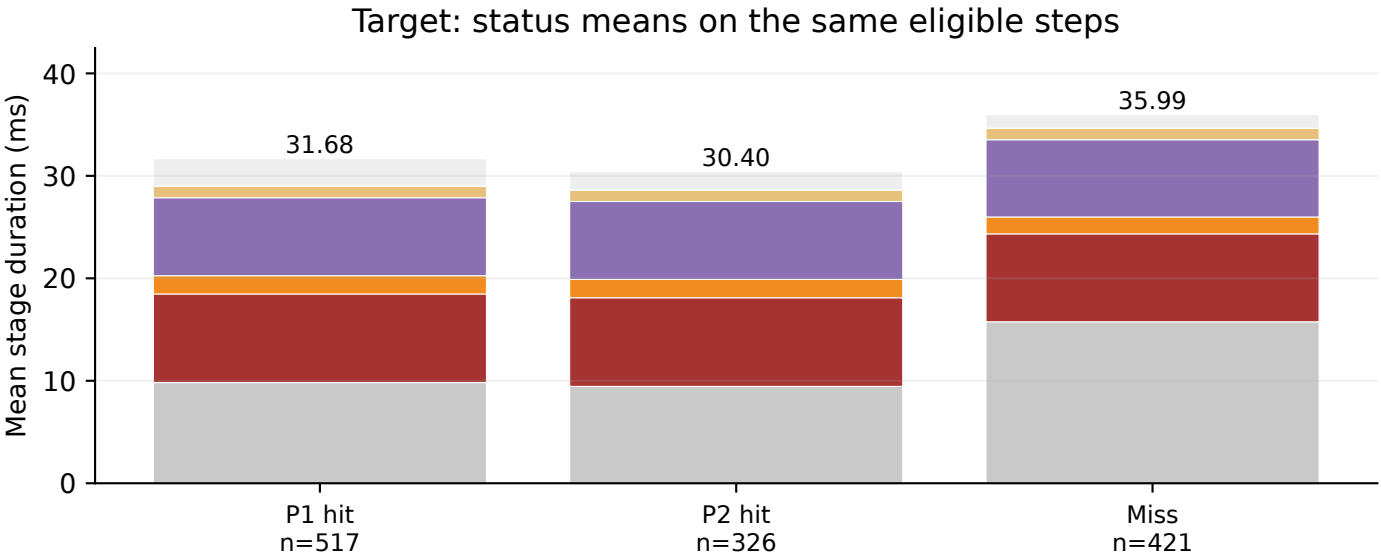
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Miss response Cache response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context Send response Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request SSD build

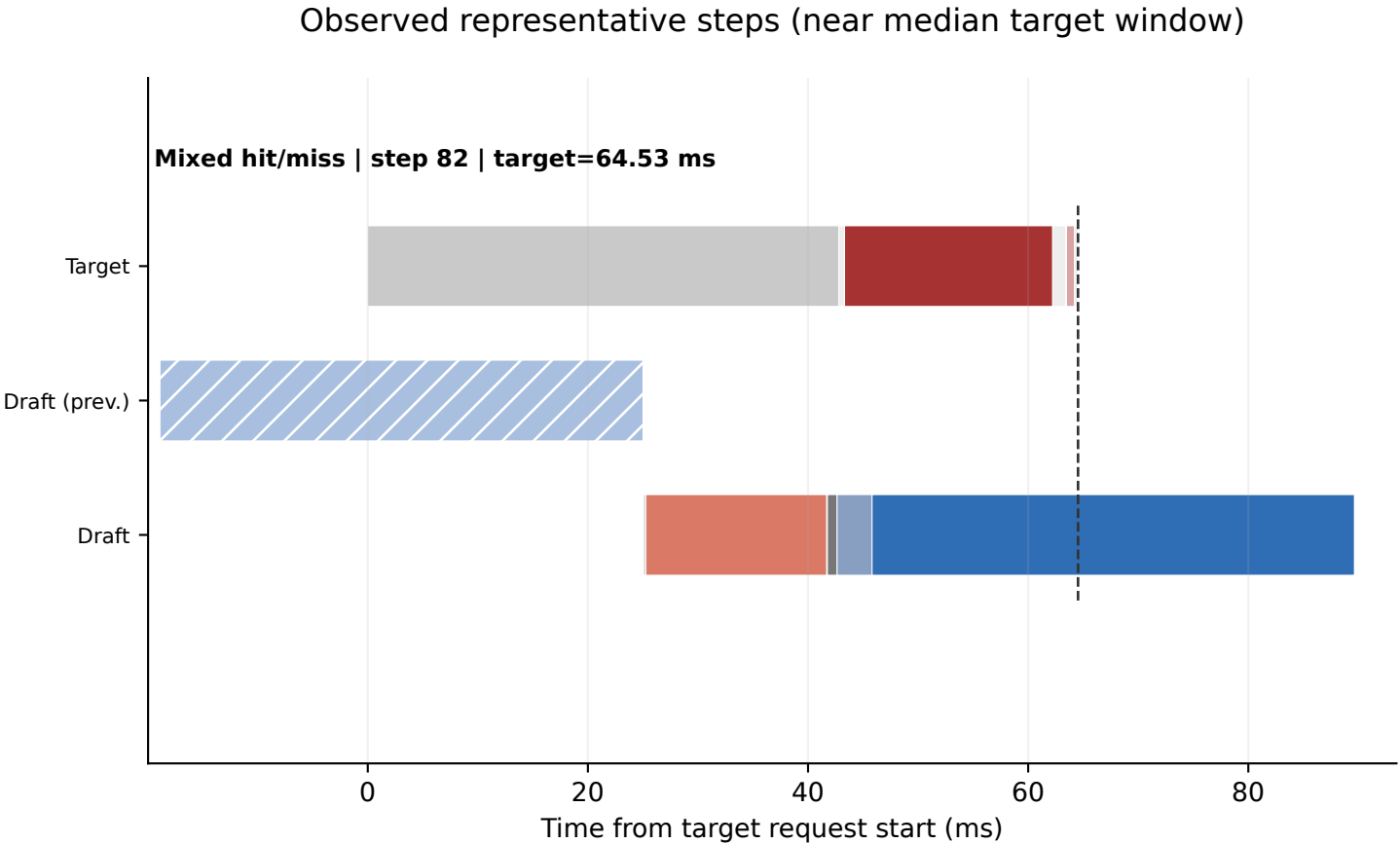
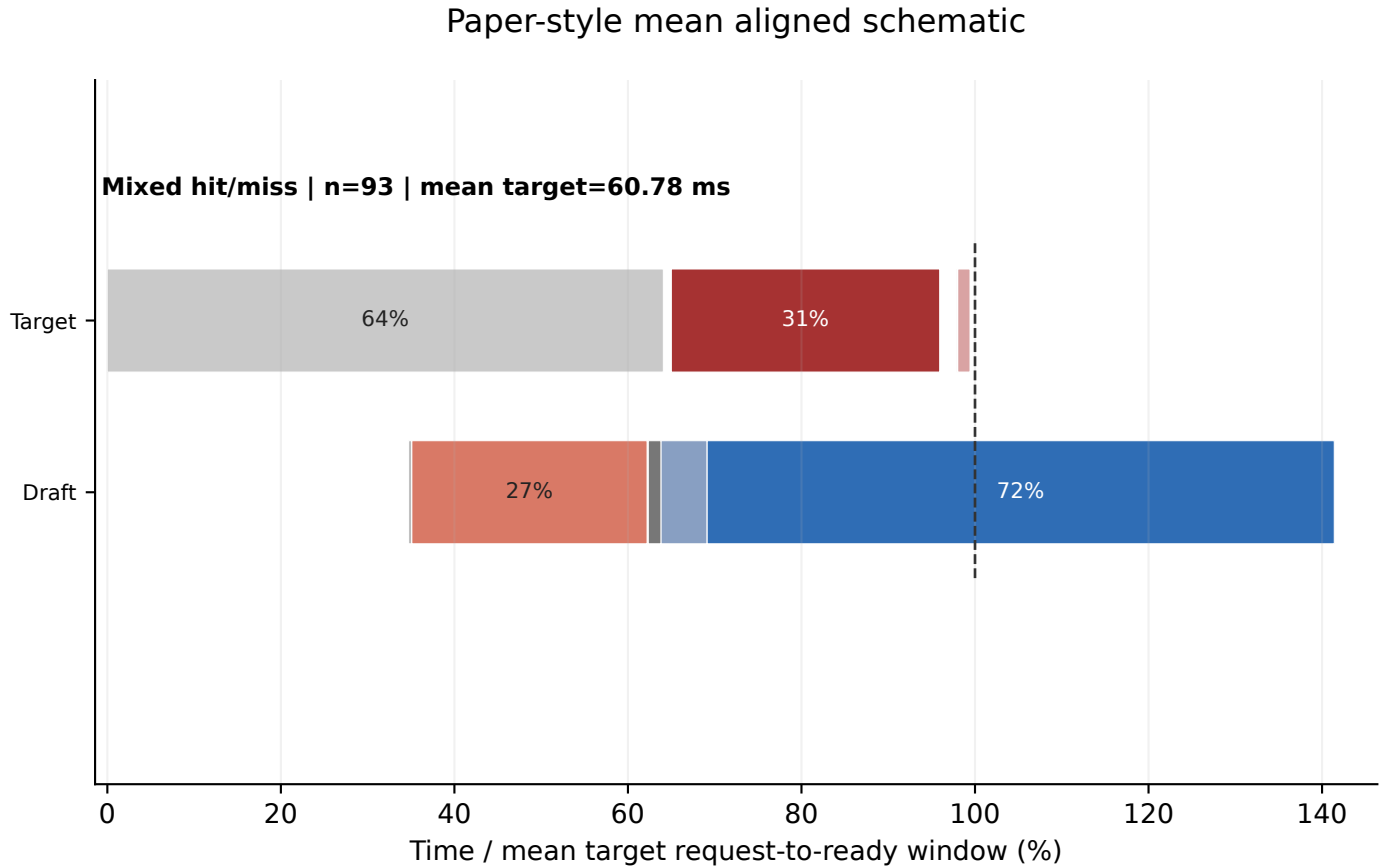
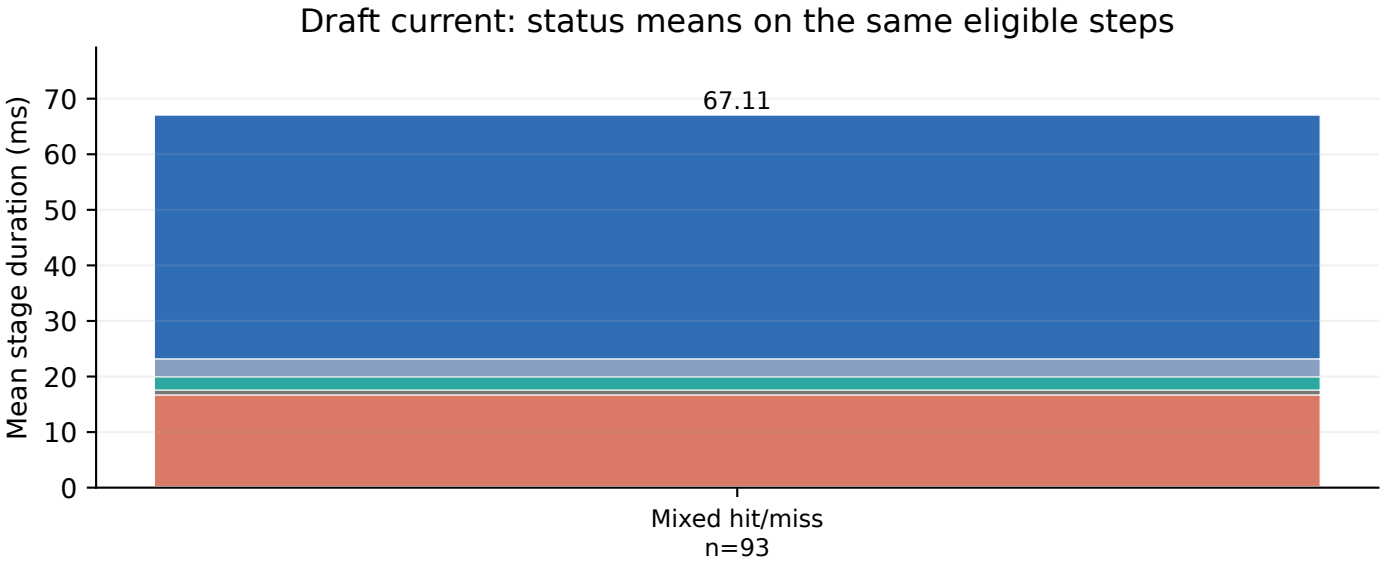
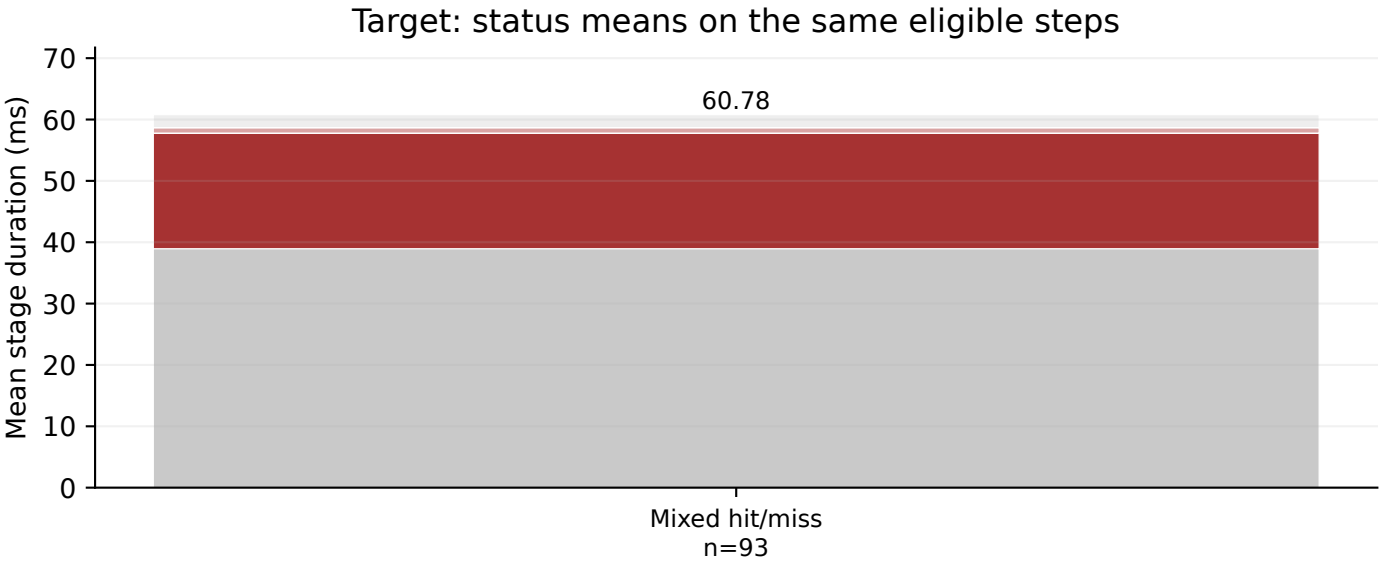
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b1_refine_e16_k6_4_r2
DUET: K1/K2=6/4, exit=16 (0-based), $\bar{C}$ =3, N1/N2=12/8, U=2, W=7, trim/stream=0/0
B=1, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 1264 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

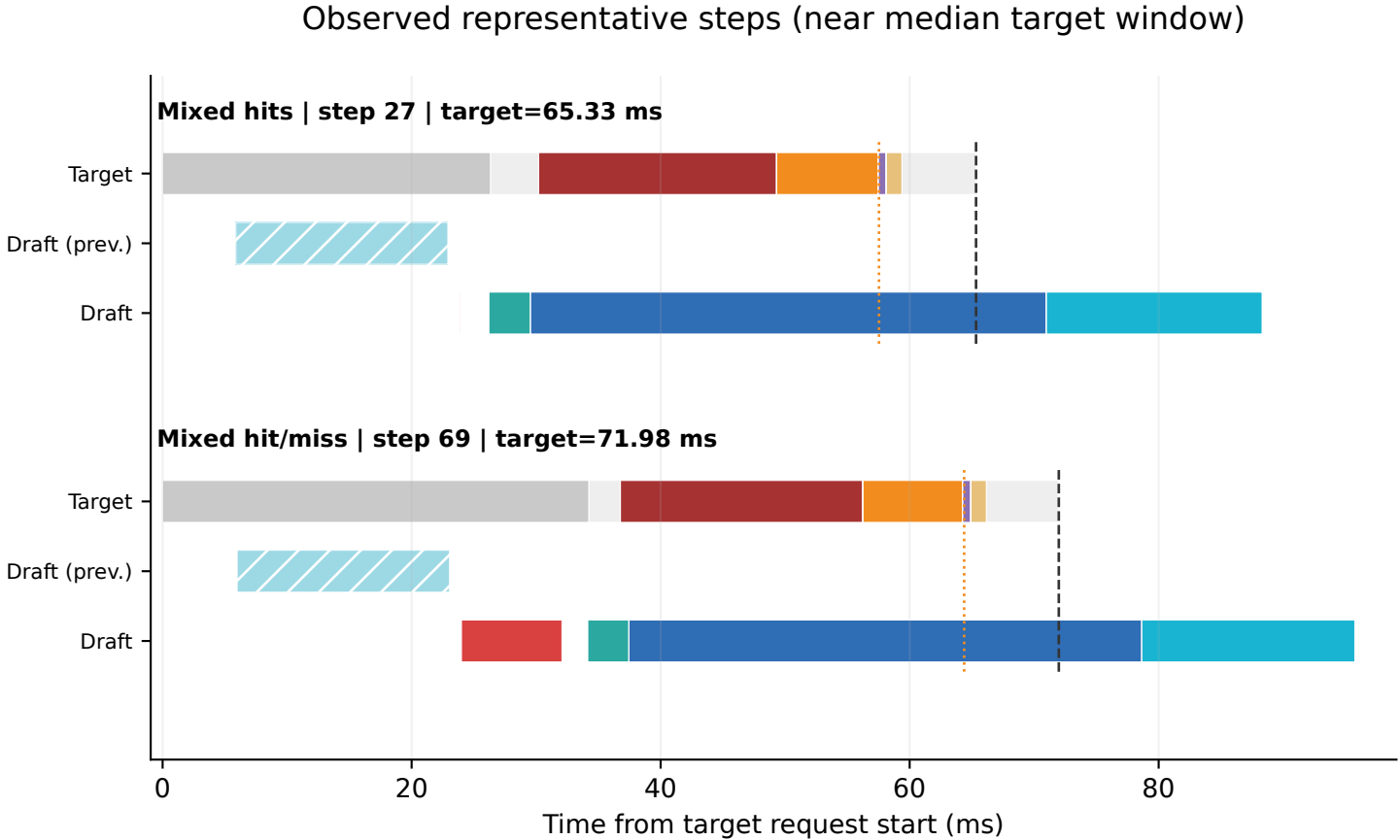
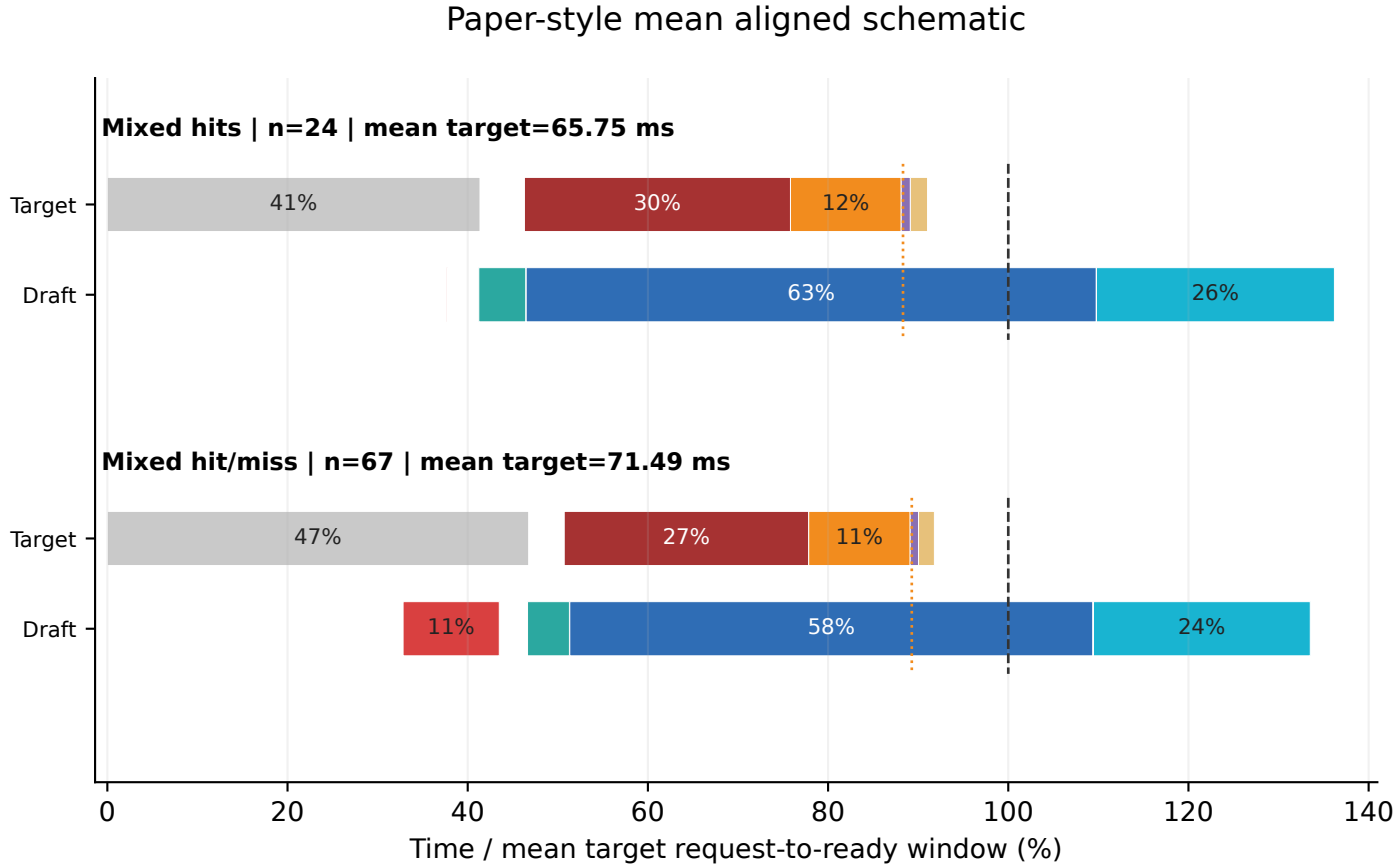
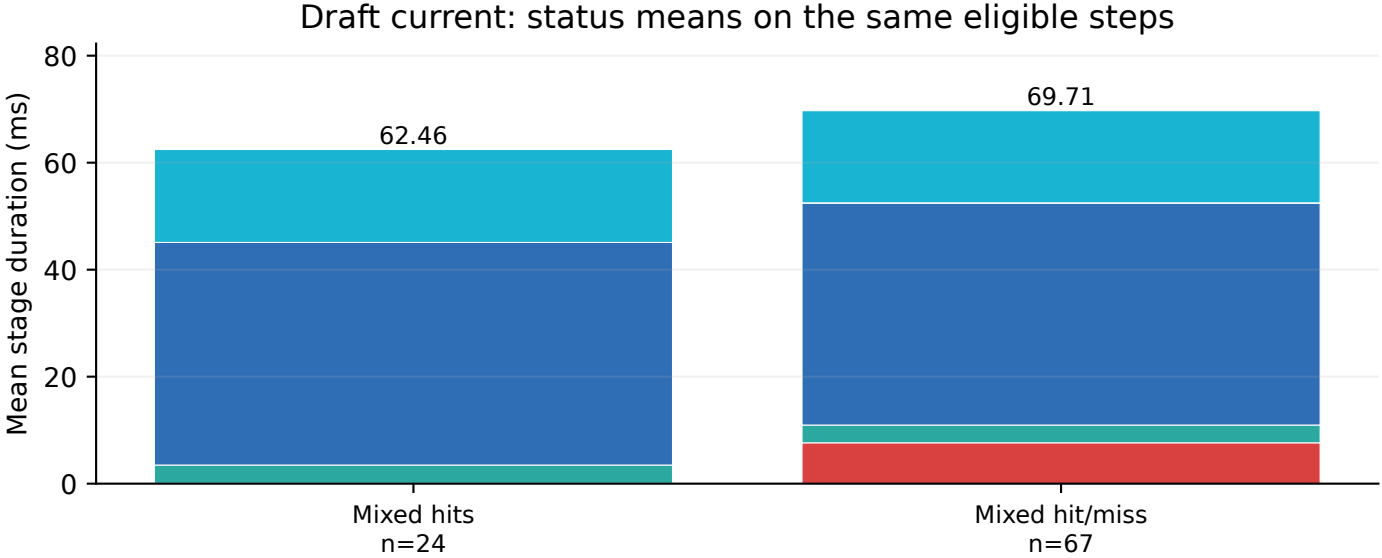
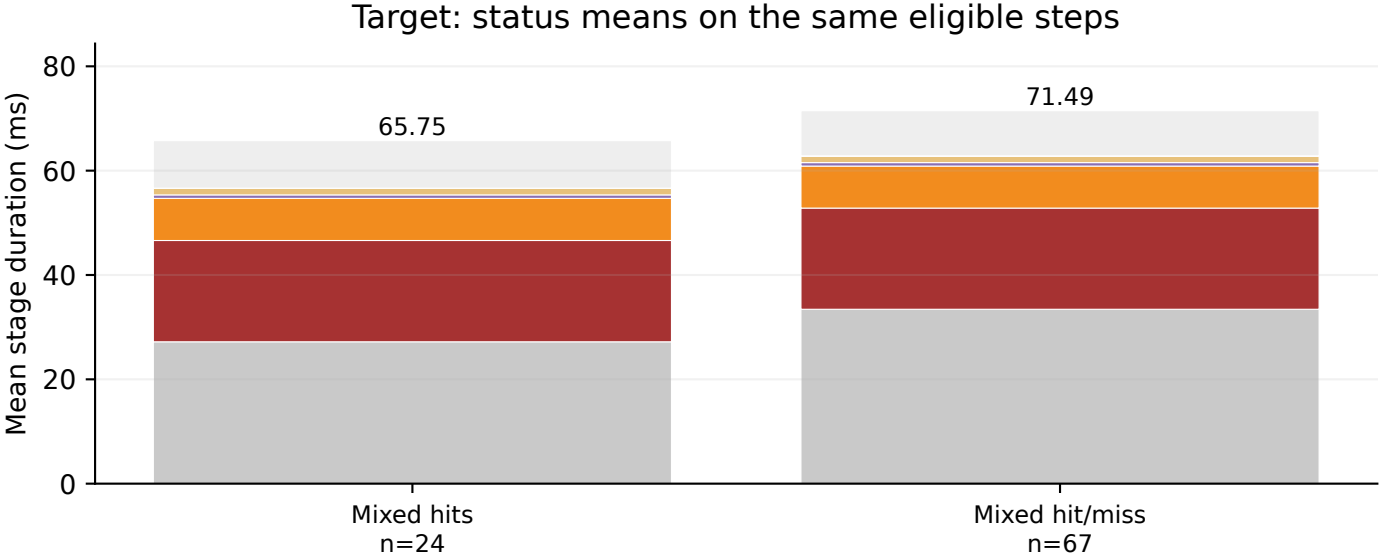
Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.



Wait / sync Accept prepare Mixed response Send response SSD decode Previous SSD decode Target ready
Target verify Postprocess Context SSD build Cache insert Previous cache insert Proxy received
Accept Unlabelled gap Request

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.

llama3_b8_combo_al_k2_3
DUET: K1/K2=6/3, exit=30 (0-based), C=3, N1/N2=12/4, U=2, W=7, trim/stream=0/0
B=8, T=0.7, 48 prompts, cap=512/96 | instrumented diagnostic; 91 eligible steps



Wait / sync Proxy / launch Postprocess Miss draft P1 Proxy wait Target ready
Target pre Final logits Unlabelled gap Context P2 Previous P2 Proxy received
Target post

Stage means are diagnostics, not final TPS. Proxy side-stream work is a separate lane, never added to the target stack.
Blank timeline gaps are retained; no common hit/miss compute cost is imposed. Cross-GPU times use recorded CPU/CUDA anchors.